

Predictive Interaction Realizability on the FR3

From a design gate to a two-axis certificate — what holds, what broke, and what it costs

Working draft, simulation only. It began as one question — is the passive-nominal split feasible? — and the answer to that (§1–§6) turned out to be the least interesting thing in it. The merged controller (§7), the fixes for what it exposed (§8), the pose study that dissolved most of those (§9), and the tension between PIR’s two axes (§10) each changed the previous answer.

How to read this. The document is written in the order the work happened, and several conclusions were later overturned by a subsequent experiment. Rather than silently rewriting them, superseded sections carry a **Corrected by §x** banner at the top and the correction is made where the newer evidence is. §11.0 tabulates every such reversal, because the pattern in them — five of seven were properties of one FR3 configuration — is itself one of the findings. If you want only the current state: §1, then §10, then §12.

1. Decision summary

Merging `phri2`’s behaviour-realization separation with `impedance_residual`’s energy-authorized residual port requires `phri2` to acquire something it does not have: a **nominal**. The proposed route is the *passive-nominal split*,

$$F_{\text{cmd}} = F_{\text{nom}} + F_r, \quad F_{\text{nom}} = -K_0 e - D_0 v,$$

in which F_{nom} is a small behaviour-agnostic spring-damper that supplies a passivity floor and a harvestable dissipation term $v^\top D_0 v$, and the realization QP optimizes only the residual F_r that the tank then authorizes. The merged Lemma 1 that this route buys is worthless unless its first precondition is satisfiable: the **anchor**

$$a_\ell = \tau_{\text{base},\ell} + J_{v,\ell}^\top F_{\text{nom},\ell}, \quad |a_\ell| \leq \bar{\tau} \quad \text{at every tick,}$$

must fit inside the derated torque envelope — and the $\alpha \rightarrow 0$ fallback it degenerates to must still hold a sustained 20 N push inside the 0.06 m workspace bound. Those two requirements pull in opposite directions. This document reports whether they have a common solution.

They do, but only under one of the two envelopes the source papers use, and only in a sliver.

Torque envelope	Source	Feasible region	Verdict
$\rho \tau_{\text{max}}, \rho = 0.28$	<code>impedance_residual</code>	0 / 900 cells	NO-GO (at this pose; see §9.2)
87/87/87/ 31.5 /12/12/12 N·m	<code>phri2</code>	33 / 900 cells , $k \in [360, 400]$ N/m, $d \in [21.3, 48.4]$ N·s/m	GO, thin

Under the $\rho = 0.28$ envelope no (K_0, D_0) works *at this pose* — and §9.2 later shows that “at this pose” is the whole of it, correcting §5.1’s original claim that the result was structural. Under `phri2`’s own envelope the split is viable at $K_0 = 380$ N/m, $D_0 = 29.1$ N·s/m, with every hard diagnostic sitting at 93–98 % of its limit (§5.2).

Three findings change what the merge can claim, and all three are load-bearing:

- The two papers’ “derated budget” stress cases are not interchangeable.** The synthesis note treats both as $\rho = 0.28$. They are not: `phri2`’s own realized torque peaks at **1.30×** the $\rho = 0.28$ cap, so `phri2`’s published FR3 benchmark does not even *run* inside `impedance_residual`’s envelope. The merge inherits whichever envelope it is written against, and the answer flips between them.
- The note’s four-diagnostic table admits false green cells.** Its row (1) is evaluated on a trajectory where F_{nom} is small, while its row (3) defines a displaced operating point the robot must actually hold. Checking each in its own place, 70 cells pass; requiring the precondition to hold on the *fallback itself* (row 1b, §4.3) leaves 33. Half the apparent region is an artifact of never asking one trajectory to satisfy both constraints.

3. **The $\alpha \rightarrow 0$ fallback transiently leaves the workspace box.** At the recommended cell the settled displacement is 55.8 mm against the 60 mm bound, but the transient peaks at **62.2 mm**. Gate on peak instead of settled displacement and the region drops from 33 cells to 6 — and the recommended cell is not among them.

The merged controller has since been implemented and run in closed loop (§7). That settles the gate’s main caveat and adds a fourth finding that outranks the other three:

4. **The split converts a hard actuator guarantee into a conditional one.** In `phri2` the QP’s torque constraint covers the *whole* command. Under the split the nominal sits outside the authorization loop — the 1 kHz servo can scale F_r but never F_{nom} — so once $|\tau_{\text{base}} + J_v^T F_{\text{nom}}| > \bar{\tau}$ nothing in the fast loop can prevent the overrun. With only 2.4 % of joint 4’s cap left as headroom, that happens at $8\times$ `impedance_residual`’s own disturbance amplitude. (§8.2’s fix, adopted since, removes it: the shipped default holds the envelope at every scale tested. §7.7 records the behaviour that motivated the fix.)

§8 root-causes all four and reports what can be done about them: findings 1–3 are one problem, not three — diagnostic 3 forces K_0 above the desired impedance’s own stiffness — and finding 4 is separate and is fixed outright by giving the servo authority over the nominal (**adopted**, §8.2). §8.5 then states the repaired guarantee, **Merged Lemma 1**, whose precondition is merely $|\tau_{\text{base}}| \leq \bar{\tau}$ — a pose property, not a design one — and checks it clause by clause. It also prices the fix: where α_{nom} fires it buys a 0.74 % overrun of a *derated* cap by spending 71 points of workspace excursion, which re-opens decision 0 on terms the simulation cannot settle.

§9 then removes most of the problem rather than trading it. All four findings were measured at one FR3 pose, and that pose is a bad one: its bias torque uses 63.6 % of joint 4’s cap before any control is applied. At a pose that does not load joint 4 the feasible region grows from 33 cells to **481**, the residual’s torque headroom from 2.4 % to **45.7 %**, and the workspace excursion under a $16\times$ disturbance *improves* from 164 mm to 59 mm. Findings 1–3 were substantially an artifact of the scenario, not of the architecture.

§10 is the one that changes what the paper claims. Aggregating every run, the two axes turn out to be *anti-correlated*: at `phri2`’s pose the feasibility axis fires and r_{auth} is exactly zero; at the recommended pose r_{auth} removes 46 % of the intended behaviour and α_r never fires at all. Both act on the same residual through the same torque budget, so a nominal stiff enough to make saturation bind leaves too small a residual to drain the tank, and vice versa. The framing that survives this is stronger than “both axes matter”: **you cannot tell in advance which axis will bind**, and one innocuous-looking joint-angle change flips it. A joint pose $\times K_0$ scan (§10.5, 96 cells) then finds exactly one operating point where both axes carry real load with every certificate intact — and §10.6 resampling it across 10 seeds shows that point keeps all its certificates in only **5 of 20** runs, always failing on Lemma 1’s precondition. The flip, by contrast, is **20/20**. So the claim the evidence supports is the flip, and the dual-axis regime is a remark rather than an exhibit.

§12 states what is left for a human.

2. Why the split is needed at all

The two unsubmitted drafts are complementary halves of “predictive impedance under saturation”:

- `impedance_residual` holds a fixed physical impedance as nominal, lets a 100 Hz MPC propose an additive residual wrench, and authorizes that residual at 1 kHz against both a joint-torque envelope and an energy tank. Its central result is that the residual port must be re-authorized *between* manager updates.
- `phri2` emits a desired contact-port acceleration $a_{id} = C_\theta x + G_\theta F_h$ from a behaviour layer, realizes it in a receding-horizon QP under a horizon-wide torque constraint, and reports the deviation as $r = r_{\text{reg}} + r_{\text{con}} + r_{\text{mod}}$ rather than hiding it in clipping.

`phri2` §VIII names its own gap in as many words: “*the realization cost does not itself imply passivity ... A dissipativity constraint on the realized port variables, paired with an energy tank or passivity observer, is a natural extension.*” That extension is `impedance_residual`’s tank.

The graft does not work naively, for a structural reason. `impedance_residual`’s tank sits on a nominal τ_I that is always-feasible and natively passive, and it authorizes only the additive residual. `phri2` **has no nominal**. Its τ_{base} is feedforward plus orientation hold plus null-space centering, with no impedance term at all; the entire interaction behaviour lives inside F_{cmd} . Two consequences follow:

1. Scaling F_{cmd} down at a fast tick degenerates not to a passive spring-damper but to **zero interaction behaviour** — gravity compensation and posture only, a free-floating arm.
2. The tank’s measured advantage comes from harvesting the nominal damping $v^\top D_I v$. `phri2` has no harvestable damping term.

The passive-nominal split exists to fix both at once, at the cost of introducing (K_0, D_0) — and with them the question this document answers.

3. The merged fast re-authorization rule

For context, the rule the split is meant to enable. The QP supplies $F_{r,k}$ every 20 ms; the following runs every 1 kHz tick ℓ ($h = 1$ ms):

1. Recompute from the current $(q, \dot{q})$: $\tau_{\text{base},\ell}$, $J_{v,\ell}$, Λ_ℓ , and $F_{\text{nom},\ell} = -K_0 e_\ell - D_0 v_\ell$.
2. **Anchor** $a_\ell = \tau_{\text{base},\ell} + J_{v,\ell}^\top F_{\text{nom},\ell}$, with the precondition $|a_\ell| \leq \bar{\tau}$.
3. **Torque scale** $\alpha_{\tau,\ell} = \text{largest } \alpha \in [0, 1] \text{ with } |a_\ell + \alpha J_{v,\ell}^\top F_{r,k}| \leq \bar{\tau} \text{ per joint.}$
4. **Energy scale** $\alpha_{E,\ell} = 1$ if $\bar{F}_{r,\ell}^\top v_\ell \leq 0$, else $\min(1, (E_\ell - E_{\min} + h v_\ell^\top D_0 v_\ell) / (h \bar{F}_{r,\ell}^\top v_\ell))$.
5. **Apply** and **ledger** as in `impedance_residual`.

Steps 1-2 are what §4-§6 gate. Steps 3-6 are implemented in `pir_controller.py` and measured in §7.

4. Method

4.1 One FR3 model, imported not rewritten

Every mass matrix, Jacobian, task-space inertia, bias torque, null-space projector, torque limit and scenario constant is imported from the existing studies (`simulation/fr3_mujoco.py` , `simulation/fr3_impedance.py` , `imp_reference/simulation/fr3_interaction_dynamics_mpc.py` , `imp_reference/simulation/run_fr3_experiments.py`). Nothing is re-derived: a second FR3 model would make this gate inconsistent with the papers it gates. The regression suite asserts the 0.06 m bound *is* `FR3MPCConfig.position_limit` rather than merely equalling it.

4.2 Scenario

`phri2` ’s FR3 benchmark: the EE holds a fixed nominal pose while a smooth sustained 20 N push along $-z$ (raised-cosine ramp, `human_force_at`) would, under naive realization, drive it through the 0.06 m workspace bound. 6 s, 1 kHz servo, 50 Hz manager. Task-space inertia at the nominal pose is $\Lambda_{zz} = 4.56$ kg; the bias torque on joint 4 is 18.96 N·m before any control is applied at all.

Two torque envelopes are carried, because the two papers use different ones:

- `rho_0.28` — $0.28 \tau_{\max}$ on every joint (24.36 N·m on joints 1-4, 3.36 N·m on joints 5-7), `impedance_residual` ’s continuous-envelope stress case.
- `derated_joint4` — stock FR3 limits with joint 4 derated $87 \rightarrow 31.5$ N·m, `phri2` ’s own torque-activation stress case.

4.3 Diagnostics

Rows 1-4 follow the synthesis note’s §13.2 table. Row 1b is added.

#	Quantity	Checks	Pass
1	$\max_t \ \tau_{\text{base}} + J_v^\top F_{\text{nom}}\ _\infty / \bar{\tau}$ on the <code>phri2</code> replay	anchor feasible	< 1
1b	the same ratio, peaked over the closed-loop $\alpha \rightarrow 0$ fallback run	anchor feasible <i>there too</i>	< 1
2	$\lambda_{\min}(K_0), \lambda_{\min}(D_0)$	passive	> 0
3	settled $\ e_z\ $ at $\alpha \rightarrow 0$ under the 20 N push	fallback holds	$\leq 0.06 \text{ m}$
4	$\max_t \ J_v^\top F_{\text{nom}}\ _\infty / \bar{\tau}$	not budget-dominating	$< \beta = 0.5$

Why row 1b exists. Rows 1 and 3 are measured at two different operating points. Row 1 replays the trajectory `phri2`’s MPC actually produced, on which the EE stays near the nominal pose, so $F_{\text{nom}} = -K_0 e$ is small whatever K_0 is. Row 3 describes the *displaced* equilibrium the arm settles into once the residual is de-authorized, where by construction $K_0 e = F_h$. Merged Lemma 1 requires $|a_\ell| \leq \bar{\tau}$ at **every** tick, and the fallback’s ticks are ticks. Row 1b closes that hole by peaking the same ratio over the fallback run itself — transient included, since Lemma 1’s precondition has no exemption for overshoot. Omitting it more than doubles the apparent feasible region (§5.2).

Rows 1 and 4 are replay diagnostics: cheap, and exact *given* that the merged controller’s total command tracks `phri2`’s. That assumption is the main caveat on this result and is stated as such in §7.

Row 3 is measured on a closed-loop MuJoCo run of the actual $\alpha \rightarrow 0$ law, $\tau = \tau_{\text{base}} + J_v^\top F_{\text{nom}}$, not from static algebra. The push hold is extended to 6 s and the displacement averaged over the final 1.5 s: at $k = 380 \text{ N/m}$ against $\Lambda_{zz} = 4.56 \text{ kg}$ the fallback spring has a 0.69 s natural period, so `phri2`’s own 2 s hold and a sub-period averaging window read an oscillation rather than an equilibrium — and biased the estimate in the flattering direction, making low-damping cells look *better* than they are.

4.4 Grid

$k \in [10, 1200] \text{ N/m}$ log-spaced, $d \in [2, 60] \text{ N}\cdot\text{s/m}$, densified through $k \in [300, 520] \text{ N/m}$ and $d \in [8, 40] \text{ N}\cdot\text{s/m}$: $36 \times 25 = 900$ cells. The note proposes $k \leq 400 \text{ N/m}$ and says to widen if the region sits at an edge. It does — row 3 alone forces $k \gtrsim 350 \text{ N/m}$ — and the densification is not cosmetic: on a 24×16 grid the region is $\sim 40 \text{ N/m}$ tall and the scan reports it **empty** (§6.3).

5. Results

5.1 `rho_0.28` : infeasible at this pose

Corrected by §9.2. This section originally called the result structural. It is not: it is a property of `phri2`’s nominal pose and push direction. At other poses `phri2`’s own controller runs inside the $\rho = 0.28$ envelope. Everything measured below is correct as measured; the word “structural” was an over-claim, and §9.2 gives the counter-example.

	value
τ_{base} alone, as a fraction of joint 4's cap	0.822
phri2 's own realized torque, as a fraction of this cap	1.300
cells passing row 1	197 / 900
cells passing row 1b	0 / 900
cells passing row 3	385 / 900
common region	0 cells
best worst-normalized use over the whole grid	1.238

Row 1b fails in **every cell**, with a grid-wide minimum of 1.144. This is not a tuning result *at this pose*. Holding a 20 N push at the fallback equilibrium requires $K_0 e = F_h$, so $J_v^\top F_{\text{nom}} = -J_v^\top F_h$ *regardless of K_0* — the gains set where the arm sits, not how hard it must push back. On this pose that costs ~ 28.4 N·m on joint 4 asymptotically (§6.2), against a 24.36 N·m cap. No (K_0, D_0) can change that.

The second row of the table is the more consequential one. phri2 's published FR3 trajectory peaks at 1.30× this envelope, so the envelope is not one the phri2 benchmark ever operated in. Carrying $\rho = 0.28$ into the merge, as the synthesis note assumes, does not merely tighten the design problem — it changes the scenario to one in which the source result does not exist.

5.2 derated_joint4 : feasible, thin

	value
τ_{base} alone, as a fraction of joint 4's cap	0.636
phri2 's own realized torque, as a fraction of this cap	1.005
cells passing row 1 / 1b / 3	577 / 702 / 385
common region	33 / 900 cells
extent	$k \in [360, 400]$ N/m, $d \in [21.3, 48.4]$ N·s/m
region if row 1b omitted	70 cells
region if row 3 gated on peak rather than settled	6 cells
region after $\beta = 0.5$	33 cells (row 4 never binds)

Recommended operating point. The note's literal criterion — minimize row 4 subject to rows 1–3 — selects $k = 360$ N/m, $d = 21.3$ N·s/m, which sits 0.4 % from violating row 1b. That is the wrong objective when the region is a sliver: row 4 is the only diagnostic the note does not hard-gate. The cell furthest from *every* hard constraint is the one to take:

$$K_0 = 380 \text{ N/m}, \quad D_0 = 29.1 \text{ N} \cdot \text{s/m}$$

diagnostic	value	limit	use
(1) replay anchor	0.977	1	98 %
(1b) fallback anchor	0.976	1	98 %
(3) settled $\ e_z\ $	55.8 mm	60 mm	93 %
(3') peak $\ e_z\ $	62.2 mm	60 mm	104 %
(4) budget fraction	0.348	$\beta = 0.5$	70 %

This is a damping ratio of $\zeta = 0.35$ at 1.45 Hz — a floor that is genuinely a floor, not a second behaviour layer.

The three-way squeeze. The overlay figure shows the region bounded below by row 3 (too soft and the fallback leaves the box), above by row 1 (too stiff and the anchor overruns joint 4 on the operating trajectory), and on the left by row 1b (too little damping and the fallback’s overshoot overruns joint 4). The note anticipated tensions (a)↔(c) and (c)↔(d). (d) never binds: row 4 ranges over 0.326–0.373 across the whole region, so β could be set anywhere above ~ 0.38 without changing a single cell. The binding third constraint is the one the note did not list — damping against the fallback transient.

5.3 Figures

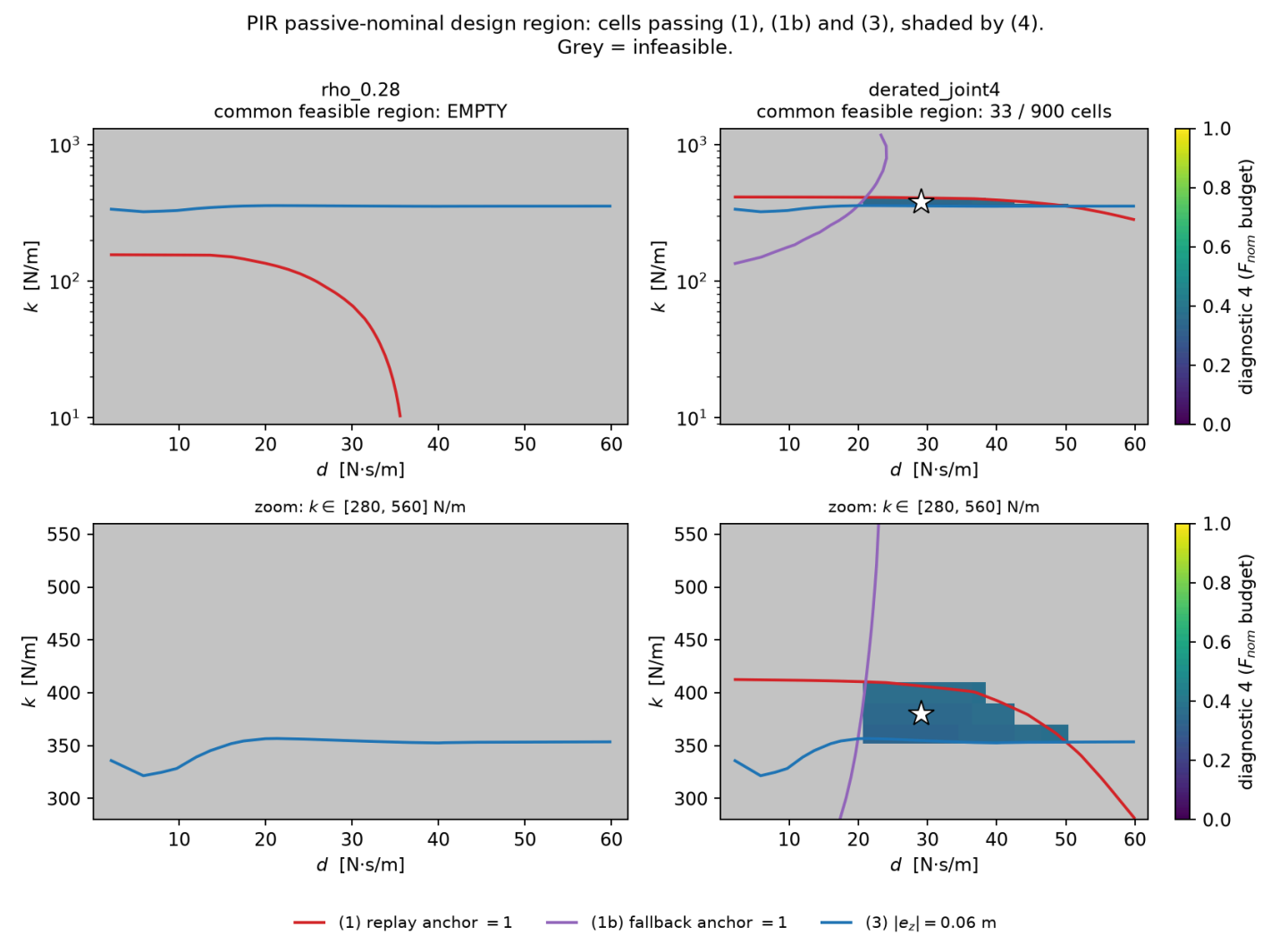


Figure 1 — the design region. Grey is infeasible. Left: the $p = 0.28$ envelope, empty everywhere. Right: `phri2`’s derated-joint-4 envelope, 33 of 900 cells, with the recommended operating point starred. The zoom row is not decoration — the region is ~ 40

N/m tall inside a 1190 N/m sweep, which is why a coarse grid reports it empty (§6.3). The three-way squeeze is visible in the lower-right panel: row (3) bounds it from below (too soft and the fallback leaves the box), row (1) from above (too stiff and the anchor overruns joint 4), row (1b) from the left (too little damping and the fallback’s overshoot overruns joint 4). The note anticipated the first two tensions; the third is the one it did not list.

file	content
<code>pir_knot_scan_diag{1,1b,3,4}_{envelope}.png</code>	the four diagnostic maps, per envelope
<code>pir_knot_scan_overlay.png</code>	composite, full range and zoomed, with the recommended cell marked
<code>pir_knot_scan.json</code>	grid, all diagnostic arrays, <code>common_region_nonempty</code> , both recommendations
<code>pir_closed_loop_{scenario}_{envelope}.png/json</code>	§7: the merged controller in closed loop
<code>pir_e0_sweep.png/json</code>	§7.4–7.5: authorization vs tracking, both tightenings active
<code>pir_rootcause.png/json</code>	§8.1: every symptom swept against K_0
<code>pir_fixes.png/json</code>	§8.3: candidate fixes scored on all four findings
<code>pir_pose_study.png/json</code>	§9.1–9.2: the pose screen and its two traps
<code>pir_knot_scan_pose_*</code>	§9.3: the full gate re-run at the better pose
<code>pir_closed_loop_pose_*</code>	§9.4: Task 2 re-run at the recommended pose
<code>pir_e0_sweep_pose_*</code>	§9.4: Task 3 re-run at the recommended pose
<code>pir_axis_tension.png/json</code>	§10: which axis fires where, aggregated over every run
<code>pir_rootcause_pose_*</code> , <code>pir_fixes_pose_*</code>	§8.4: §8 re-run at the recommended pose
<code>pir_joint_scan.png/json</code>	§10.5: the joint pose $\times K_0$ scan and its box-allowance sensitivity
<code>pir_robustness.png/json</code>	§10.6: decision 3’s two formulations across 10 seeds $\times$ 2 profiles
<code>pir_lemma_check.json</code>	§8.5: Merged Lemma 1’ checked clause by clause

6. Verification

A NO-GO on one envelope and a 33-cell GO on the other are both strong enough claims to be attacked separately (`pir_verify.py` , `results/pir_verify.json`).

6.1 Check A — equilibrium without integration

Rows 1b and 3 are read off a settled simulation, and “settled” is a judgement call for a lightly damped spring. Check A instead solves $\ddot{q}(q) = 0$ directly on MuJoCo’s own forward dynamics — no time integration at all — and compares. Across four cells the two methods agree to within **6.9 %** worst case (1.2 % at the well-damped end), all root-finds converged, all simulations flagged settled. The residual gap is the null-space posture spring, which makes the settled configuration mildly path-dependent; it is smaller than the 7 % margin the recommended cell carries on row 3, but not by much, which is part of why §7 calls that margin thin rather than adequate.

6.2 Check B — the fallback anchor does not depend on the gains

Sweeping k over $[100, 1200]$ N/m at the root-found equilibrium:

k [N/m]	$\ e_z\ $ [m]	$k \ e_z\ $ [N]	joint-4 anchor [N·m]
100	0.198	19.83	25.29
200	0.101	20.12	27.87
400	0.050	20.06	28.34
800	0.025	19.98	28.40
1200	0.017	19.95	28.39

$k \|e_z\|$ recovers the 20 N push to within 1 % across a $12\times$ range of k , confirming $K_0 e = F_h$ and therefore that the equilibrium anchor is a property of the push and the pose, not of the gains. It converges to ~ 28.4 N·m — comfortably above the 24.36 N·m $\rho = 0.28$ cap and comfortably below the 31.5 N·m derated-joint-4 cap. That single number is why the two envelopes give opposite answers.

As a side result, the $k = 200$ row independently reproduces `phri2`’s own reported figure that a pure $K_d = 200$ impedance gives 0.10 m of static displacement against the 0.06 m bound.

6.3 Check C — grid refinement

Re-scanning $k \in [300, 500] \times d \in [8, 40]$ on a *linear* grid whose points deliberately do not coincide with the main scan’s:

envelope	region	best worst-normalized use
<code>rho_0.28</code>	empty (0 / 99)	1.244
<code>derated_joint4</code>	14 / 99	0.977

Two things follow. The `derated_joint4` verdict and its best cell are reproduced independently (0.977 against the main scan’s 0.977). And the refinement is *necessary*: the original 24×16 grid reported `derated_joint4` empty. The region is ~ 40 N/m tall inside a sweep spanning 1190 N/m, and a grid that samples it at two points misses it. Any re-run of this gate at a different pose, push direction or envelope must re-densify before trusting an empty answer.

6.4 Regression suite

`test_pir_knot_scan.py`, 12 tests, all passing. They pin the constants to their sources, check the anchor arithmetic against the per-tick matrix product, assert the zero-gain limit reduces the anchor to τ_{base} , verify the $K_0 e = F_h$ relation on the plant, check the averaging window exceeds the fallback spring’s natural period, and assert the published grid stays dense enough through the feasible band.

7. The merged controller, implemented

Task 1’s gate was a *feasibility* argument; §7 of its first draft owed a closed-loop measurement. `pir_controller.py` implements the merged rule of §3 and `run_pir_closed_loop.py` runs it on the FR3.

7.1 What the merge actually required

Three things changed relative to grafting the tank onto `phri2`, and all three live in the QP:

- The nominal is closed-loop inside the horizon.** F_{nom} is state feedback, so the prediction runs on $A - BG_0$ with $G_0 = [K_0 \ D_0]$. Freezing F_{nom} across the horizon would let the QP plan against a nominal its own plan invalidates.
- The torque rows became state-dependent.** In `phri2`, $\tau_k = \tau_{\text{base}} + J_v^\top F_{\text{cmd},k}$ is affine in the decision variable alone. Here $\tau_k = \tau_{\text{base}} + J_v^\top (-G_0 x_k + F_{r,k})$, and x_k is itself affine in every earlier residual, so each torque row carries the accumulated nominal reaction.

3. **The behaviour input and the disturbance model had to be split.** `phri2` has one force signal driving both $a_{id} = Cx + GF_h$ and the plant prediction, which is sound only when every external force is interaction force. Running under a disturbance, they are different signals. `control()` now takes `force_forecast` and `behaviour_forecast` separately, defaulting to equal — which recovers `phri2` exactly, and is asserted to. **This does not solve force misclassification**; deciding which measured force goes in which channel *is* that unsolved problem. It moves the assumption into a signature instead of hiding it in an equality.

One incidental gain: `phri2` 's QP has no principled infeasibility fallback and drops to its clipped reactive law. Here $F_r = 0$ is exactly the $\alpha \rightarrow 0$ passive nominal the Task 1 gate certified, so an infeasible solve degrades to a *certified* behaviour rather than an improvised one. (No solve was infeasible in any run reported here: 0 / 300 per run.)

7.2 Rows (1) and (4), re-measured in closed loop

This is the debt the first draft recorded. At $K_0 = 380$ N/m, $D_0 = 29.1$ N·s/m on `phri2` 's 20 N push:

	replay estimate (Task 1)	merged closed loop	gap
row (1) anchor ratio	0.9771	0.9762	0.09 %
row (4) budget ratio	0.3478	0.3474	0.11 %

The replay approximation holds. The gate's 2 % margin on row (1) survives, and the GO verdict does not need revisiting on this scenario. A regression test pins the two together at 1 % so the gate cannot silently drift from the controller it gates.

7.3 Merged Lemma 1 and the four-term residual

Across every run in §7.4–7.5 (36 sweep points plus 8 closed-loop runs), with the precondition holding, the conclusion held: $|\tau_\ell| \leq \bar{\tau}$ at every one of 6000 ticks, with α_τ saturating the envelope exactly ($\max |\tau|/\bar{\tau} = 1.0000$) rather than overshooting it.

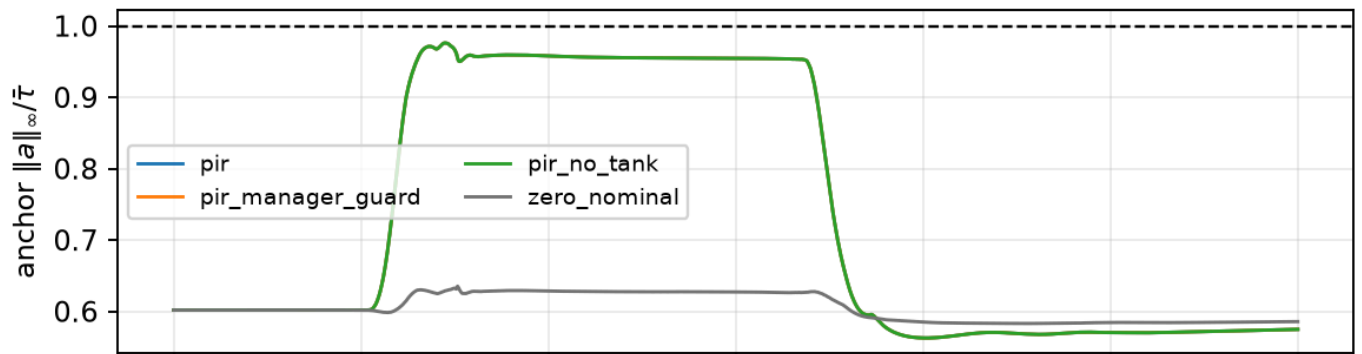
The four-term residual $r = r_{\text{reg}} + r_{\text{con}} + r_{\text{mod}} + r_{\text{auth}}$ closes **to machine precision** (4.4×10^{-16}) on every QP tick, with r_{con} split into its plan-level and execution-level halves:

$$r_{\text{con}} = \underbrace{a_{\text{con}} - a_{\text{unc}}}_{\text{QP, 50 Hz}} + \underbrace{\Lambda^{-1}(\alpha_\tau - 1)F_{r,k}}_{\text{servo, 1 kHz}}, \quad r_{\text{auth}} = \Lambda^{-1}(\alpha_E - 1)\bar{F}_{r,\ell}.$$

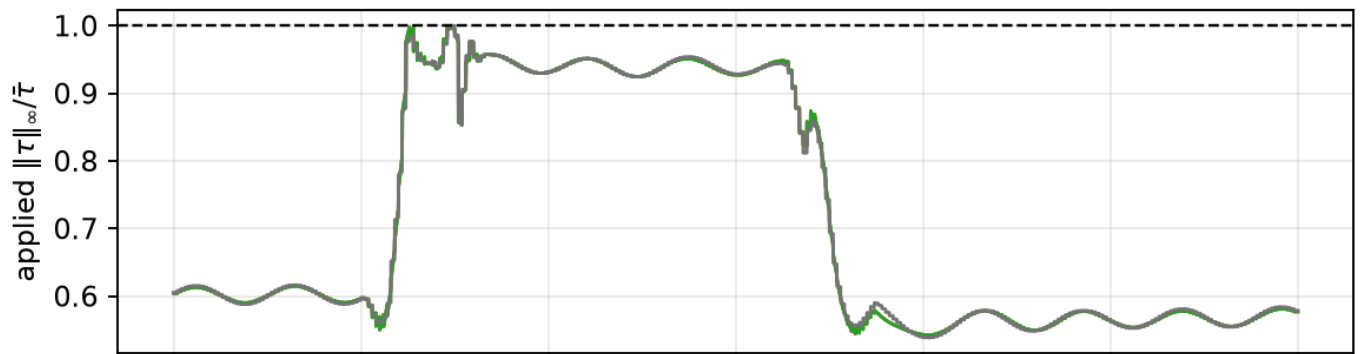
The closure is an identity, not a fit: the two servo terms telescope to $\Lambda^{-1}(F_{r,\ell} - F_{r,k})$, which is exactly the gap between what the QP proposed and what was applied. r_{auth} is identically zero whenever authorization never fires — it measures intervention, it is not an always-on correction.

PIR merged controller, scenario 'merged', envelope 'derated_joint4'

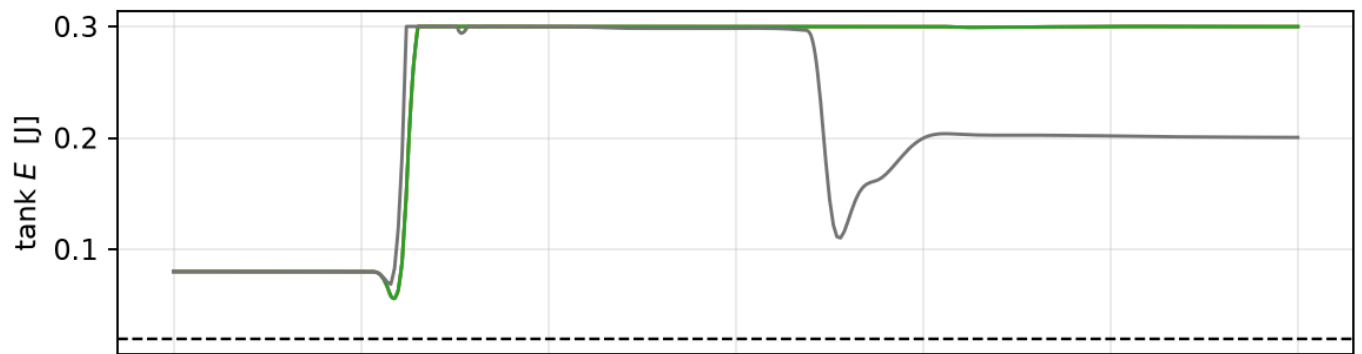
Merged Lemma 1's precondition (row 1), measured in closed loop



Merged Lemma 1's conclusion



energy tank vs its floor $E_{\min} = 0.02$ J



workspace excursion on the push axis vs the 0.06 m bound

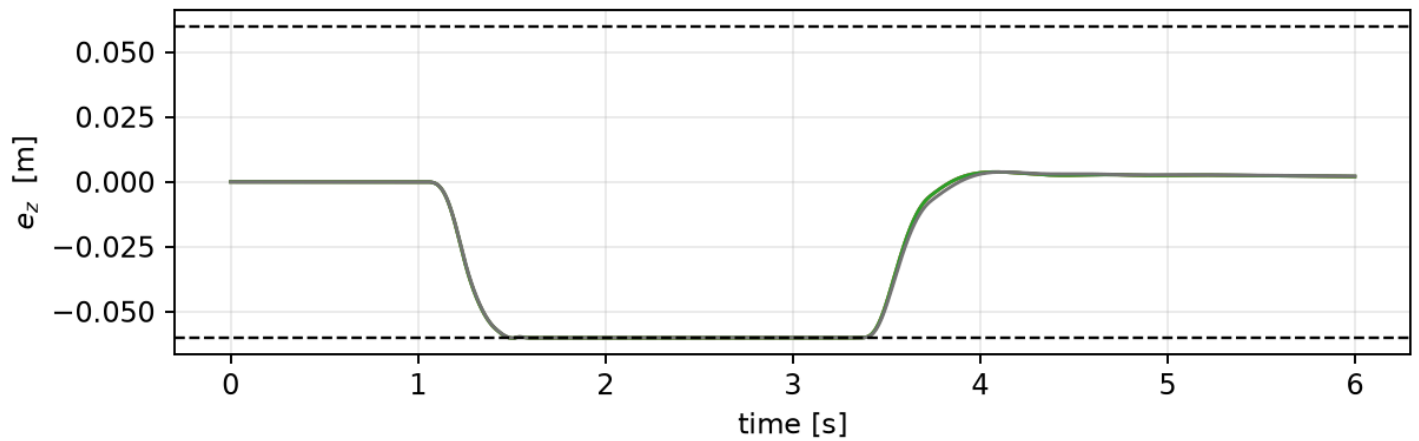


Figure 2 — the merged controller in closed loop, on the merged scenario at the recommended operating point. Top: Lemma 1’s precondition, the anchor ratio, sitting at 0.976 for the three PIR variants (they coincide here because authorization never fires at disturbance scale 1) against 0.635 for `zero_nominal`. That gap *is* \$7.6’s headroom collapse. Second panel: the conclusion — applied torque pinned at the envelope, never through it. Third: the tank, comfortably above its floor in this scenario, which is exactly why \$7.4 has to stress it deliberately. Bottom: the workspace excursion, slightly past the slack-relaxed 0.06 m box for every variant.

7.4 The passivity axis: `impedance_residual` ’s result transfers

Neither source benchmark exercises both axes. `phri2` ’s push is monotone and largely dissipative: the nominal harvests $v^\top D_0 v$ faster than the residual drains the tank, and $\alpha_E \equiv 1$ throughout. So the sweep runs on a **merged scenario** — that push plus `impedance_residual` ’s own rejectable disturbance (0.9/1.4/1.9 Hz sinusoids and a 12 N pulse placed deliberately between two 50 Hz manager ticks) — and sweeps E_0 toward its floor and the disturbance amplitude upward. 36 points, three variants:

variant	tank floor held	breached ticks	worst tank
<code>pir</code> (1 kHz re-authorization)	12 / 12	0	0.0200 J = $E_{\min}$
<code>pir_manager_guard</code> (20 ms, held)	4 / 12	6 189	−1.285 J
<code>pir_no_tank</code>	4 / 12	13 050	−28.43 J

(Re-measured after decision 0 made α_{nom} the default, which is why the breach counts are larger than the pre-decision-0 run: keeping the torque envelope legal at 16× disturbance leaves more energy for the ledger to have to account for. `pir` still holds the floor everywhere.)

`impedance_residual` ’s central result transfers to the merged port. Fast re-authorization holds the floor in every stressed configuration — and holds it *with equality*, $\min E = E_{\min}$ exactly, which is what the α_E construction guarantees. The manager-rate guard tracks it closely and breaches anyway, in 8 of the 12 configurations where the tank is loaded at all. The 20 ms staleness is the whole difference.

Authorization goes from silent to active as either axis is loaded: 0 % → 1.9 % of ticks as $E_0 \rightarrow E_{\min}$, and 0 % → **73.4 %** of ticks as the disturbance scales 1 → 12. (That last figure counts any of the three scalings intervening, so past 4× it is dominated by α_{nom} , decision 0’s fix, doing the work the anchor overrun used to do unchecked.)

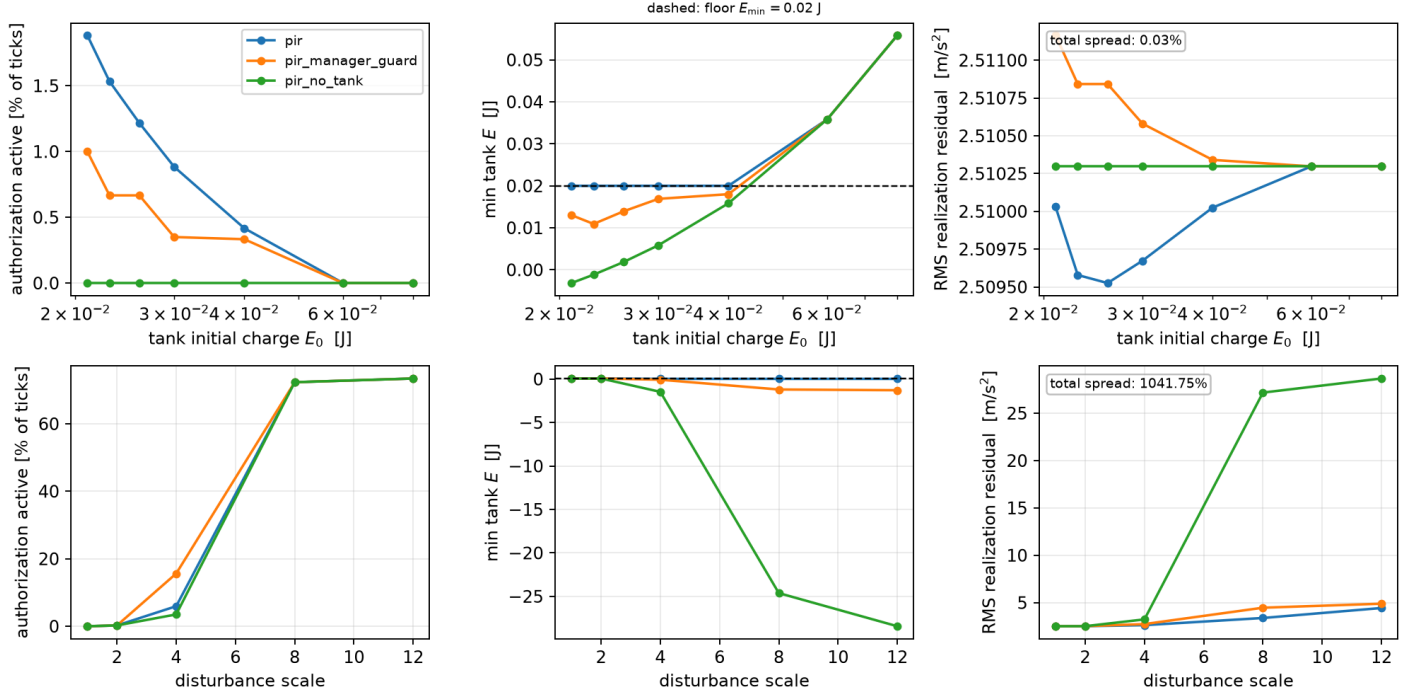


Figure 3 — authorization vs tracking, both tightenings active. Top row sweeps the tank’s initial charge toward its floor; bottom row sweeps the disturbance amplitude. The middle column is the result: **pir** (blue) sits *exactly* on $E_{\min}$, which is what the α_E construction guarantees; **pir_manager_guard** (orange) dips below it to -1.29 J; **pir_no_tank** (green) reaches -28.4 J. The right column carries its own spread annotation because matplotlib autoscales it — on the E_0 axis the total variation is 0.03 %, i.e. flat (§7.5), and only on the disturbance axis does a real trade appear.

7.5 The authorization-vs-tracking trade looks flat here

Corrected by §9.4. This subsection originally concluded that the merged E_0 trade is flatter than *impedance_residual*’s, against the synthesis note’s prediction that it would be steeper. That conclusion was a pose artifact. At the recommended pose the same sweep gives a **7.7 %** spread and the note’s prediction is confirmed. The measurement below is correct at this pose; the generalisation from it was not.

The synthesis note predicts the E_0 curve gets steeper after the merge, because the two tightenings compound in series. Here it is flat: over the whole E_0 sweep, RMS realization residual varies by **0.03 %** ($2.5095 \rightarrow 2.5103$ m/s²), against *impedance_residual*’s own $21.49 \rightarrow 15.98$ mm swing on its unmerged nominal.

The reason is the §7.6 boundary. The passive nominal has already absorbed the authority the tank would otherwise take away: de-authorizing a residual that is 16 % of the command costs almost nothing, because the other 84 % is a PD law the tank has no say over. The passivity axis is *cheap* here precisely because it is *weak* — and §9.4 shows that when the residual is given room, the cost appears exactly as the note said it would.

On the disturbance axis, where the residual does matter, the trade appears even at this pose: RMS spans $2.510 \rightarrow 4.434$ m/s² (77 %).

7.6 The nominal dominates the command, and eats the headroom

At the only (K_0, D_0) the gate certifies:

	pir	zero_nominal ($K_0 = D_0 = 0$)
$\ F_{\text{nom}}\ $ RMS	14.05 N	0 N
$\ F_r\ $ RMS	2.11 N	13.48 N
residual's share of the command	15.6 %	100 %
anchor headroom left for F_r	2.4 % of the cap	36.4 %

The note’s §7 predicted this qualitatively — “ K_0/D_0 large $\Rightarrow$ floor eats budget, a *new* double-tightening.” The measurement is worse than the phrasing suggests. $K_0 = 380$ N/m is not a small passivity floor sitting under the behaviour: it is **stiffer than the desired impedance** ($K_d = 200$ N/m), so the residual’s job is partly to make the arm *softer* than the floor. The merged controller at its certified operating point is a stiff PD with a 16 % predictive correction, and the residual’s torque headroom has collapsed **15-fold**, from 36.4 % of joint 4’s cap to 2.4 %.

7.7 Where the guarantee breaks

Measured before decision 0, and fixed by it. The table below is the controller without α_{nom} — the variant now kept as `pir_no_nominal_auth`. It is what motivated §8.2’s fix. The shipped default holds $\max|\tau|/\bar{\tau} = 1.0000$ at every scale tested, including the two that break here, so **the failure below is no longer the behaviour of the controller this document recommends**. It is retained because the precondition it exposes still fails at those scales — α_{nom} keeps the envelope legal, it does not make the anchor feasible — and because §8.4 shows the fix is inert at the pose §9 recommends, so the failure mode is dormant rather than gone.

That 2.4 % headroom is what §1’s fourth finding cashes out. Sweeping the disturbance amplitude, without α_{nom} :

disturbance scale	max anchor ratio	ticks with infeasible anchor	max $\ \tau\ /\bar{\tau}$
4	0.9888	0 / 6000	1.0000
8	1.0074	39 / 6000	1.0074
12	1.0629	558 / 6000	1.0629

Two things to read here. First, the applied-torque overrun **equals the anchor overrun exactly** at both failing scales. That is not a coincidence: when the precondition fails, α_τ has already gone to zero and the fast layer has *no remaining authority* — the overrun is entirely the nominal’s, and scaling the residual cannot touch it. Second, removing the tank as well changes nothing (1.0074 at scale 8 either way), confirming this is a property of the nominal and has nothing to do with the passivity axis.

So Merged Lemma 1 is sound and its precondition is doing real work: it is not a formality to be discharged once at design time. `phri2` guarantees the torque envelope unconditionally, because its QP owns the whole command. PIR guarantees it *conditional on the anchor being feasible*, and the Task 1 gate certified that condition against exactly one scenario. The condition does not survive an 8× disturbance. **This was the most important thing the implementation found**, and it is a cost of the split the synthesis note did not anticipate — though §8.2 fixes the consequence and §9 shows the pose largely removes the exposure, so §10 rather than this section is what ends up changing the paper’s claim.

8. Root cause, and what fixes it

Four findings is three too many if they share a cause. They do.

8.1 One constraint drives findings 1-3

Diagnostic 3 requires the $\alpha \rightarrow 0$ fallback to hold the whole 20 N push inside 0.06 m, i.e. $K_0 \geq |F_h|/0.06 = 333$ N/m. The behaviour the controller is supposed to render has $K_d = 200$ N/m. **The gate therefore forces the passivity floor to be**

stiffer than the behaviour it sits under, and everything else follows. Sweeping K_0 at fixed damping ratio $\zeta = 0.35$

(pir_rootcause.py):

K_0 [N/m]	fallback $\ e_z\ $	(3) passes	anchor headroom	residual share	largest safe disturbance	E_0 trade
100	236 mm	no	26.4 %	72.6 %	12×	0.0068
150	154 mm	no	22.1 %	59.4 %	8×	0.0071
200 = K_d	112 mm	no	17.9 %	46.4 %	8×	0.0125
250	88 mm	no	13.7 %	34.1 %	8×	0.0358
300	72 mm	no	9.5 %	23.1 %	8×	0.0057
380 = certified	56 mm	yes	2.4 %	17.1 %	4×	0.0020
460	46 mm	yes	−1.2 %	39.1 %	0×	0.0016
600	34 mm	yes	−5.3 %	48.4 %	0×	0.0010

Every symptom is monotone in K_0 , every one of them gets worse as K_0 rises, and diagnostic 3 is the only diagnostic that wants K_0 large. At $K_0 = 460$ the headroom is already negative — the precondition fails at the *nominal* disturbance. The certified band is thin from both sides.

Is diagnostic 3's premise right? It asks the passivity *floor* alone to satisfy a workspace bound that the desired *behaviour* does not satisfy either: phri2 's own $K_d = 200$ gives 0.10 m of static displacement against the 0.06 m bound (reproduced here at 112 mm, §6.2), and phri2 meets the bound through its **predictive layer**, not through its impedance. Requiring more of the floor than of the behaviour is a defensible safety choice, but it is a *choice*, not a physical constraint — and \$8.3 prices it.

Root cause: every symptom is monotone in K_0 , and only diagnostic (3) wants K_0 large

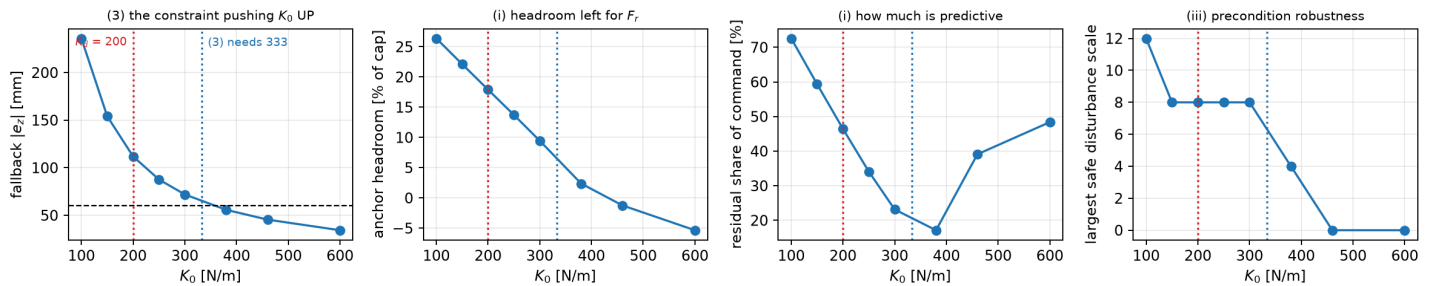


Figure 4 — the root cause. Red dotted line: the desired impedance's own $K_d = 200$ N/m. Blue dotted line: the 333 N/m diagnostic 3 demands. The certified operating point sits to the right of both, which is why the headroom and the precondition robustness are near their worst there. Only the leftmost panel improves as K_0 rises.

8.2 Finding 4 is separate, and is fixed outright

Finding 4 — the fast layer has no authority over the nominal — is not a consequence of K_0 . It is structural: α_r scales F_r only. The fix is to give the servo an α_{nom} as well, the largest scale keeping the anchor itself inside the box. Since $\alpha_{\text{nom}} = 0$ recovers τ_{base} , **the torque guarantee becomes unconditional whenever τ_{base} alone fits** — a far weaker and checkable condition (0.636 of the cap here) than “the anchor fits at every future tick”.

It is not free, and the first attempt traded one guarantee for the other. A time-varying spring gain is an energy term, and only one direction is dangerous: *softening* releases stored energy (the storage function's $\frac{1}{2}\dot{\alpha} e^\top K_0 e$ term goes negative, which helps), while *re-stiffening* is an injection — at fixed e the robot suddenly pushes back harder without the human having done the work to store

it. Charging re-stiffening to the tank and crediting softening nothing, unrestricted α_{nom} buys the torque envelope and **loses the tank floor** (−1.06 J at 16×): a transient repeatedly re-buys stiffness the port has not earned.

Forbidding the rise fixes it. With α_{nom} **monotone non-increasing**, both guarantees hold together out to 16× the source disturbance — against 4× for the certified baseline. The cost is a floor that does not recover its stiffness within an episode; a deployed system would reset it per contact, which this benchmark does not exercise.

8.3 Candidates, scored on all four findings

pir_fixes.py , all on the merged scenario, all against the same probes:

candidate	fallback	(3)	headroom	residual share	torque OK to	+ tank floor to	E_0 trade	max $\ e_z\ $ at 12×
certified $K_0 = 380$	56 mm	✓	2.4 %	17.1 %	4×	4×	0.0020	79 mm
soft_nominal $K_0 = K_d$	110 mm	✗	16.3 %	46.0 %	8×	8×	0.0070	—
anisotropic	56 mm	✓	2.2 %	20.0 %	2×	2×	0.0003	—
nominal_auth	56 mm	✓	2.4 %	17.1 %	16×	6×	0.0020	—
nominal_auth_mono	56 mm	✓	2.4 %	17.1 %	16×	16×	0.0020	—
soft_plus_auth	110 mm	✗	16.3 %	46.0 %	16×	8×	0.0070	—
soft_plus_mono	110 mm	✗	16.3 %	46.0 %	16×	16×	0.0070	160 mm

Candidate fixes for the three Section 7 findings

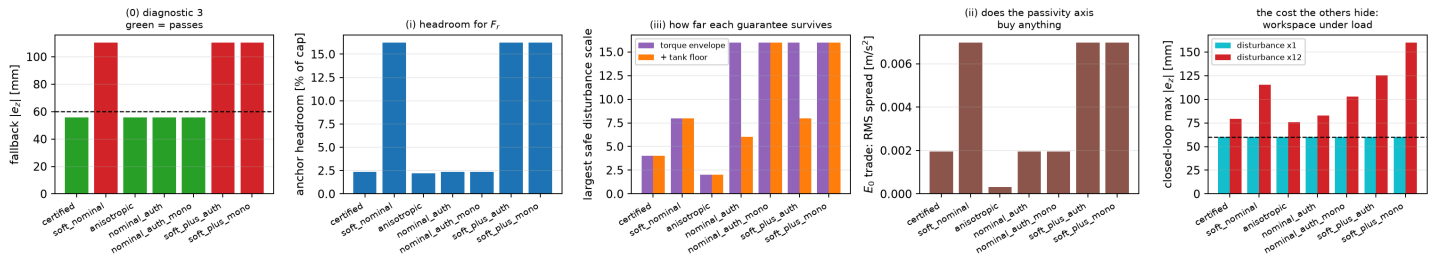


Figure 5 — candidate fixes. The rightmost panel is the one that changes the recommendation: at nominal load every candidate holds the workspace box identically, and only under a 12× disturbance does the soft nominal’s real cost appear. Without that panel `soft_nominal` looks free.

Three things to take from this.

nominal_auth_mono is a pure win and should be adopted. It quadruples the range over which *both* guarantees survive, costs nothing on any other metric, changes nothing at all while the anchor fits (asserted by test), and keeps diagnostic 3. There is no argument against it in these measurements.

The anisotropic refinement — the plan’s own deferred “later refinement” — does not work here, and slightly hurts. Headroom falls to 2.2 % and the safe disturbance range halves to 2×. The reason is visible in the trajectory: the displacement is 8:1 dominated by the push axis (max $\|e\| = [7.5, 0.3, 60.3]$ mm), so softening the off-axis gains barely reduces $\|J_v^T F_{\text{nom}}\|_\infty$ while removing the off-axis restoring force that was helping keep the arm near the pose. A negative result, but a clean one: the obvious lever is the wrong lever.

Lowering K_0 is not free, and the fallback bound is not the only thing it buys. At nominal load every candidate holds the box identically (60.2 mm), which makes `soft_nominal` look costless. Under load it is not: at 12× disturbance the certified nominal holds 79 mm while `soft_plus_mono` reaches **160 mm**, and its RMS realization residual is worse too (5.60 vs 4.03 m/s²). So the real trade for findings 1–3 is not “56 mm vs 110 mm in a fallback that may never happen” — it is **workspace containment under disturbance**, which is a live property. That is a genuine engineering decision, and it is the human’s to make, not the scan’s.

8.4 Re-run at the recommended pose: the fixes solve a problem the pose removes

§8.1–8.3 were measured at `phri2`’s pose. Re-run at the recommended one (`pir_rootcause.py --pose` , `pir_fixes.py --pose`):

K_0	headroom (<code>phri2</code>)	headroom (rec.)	safe disturbance (<code>phri2</code>)	safe disturbance (rec.)
100	26.4 %	55.8 %	12×	12×
250	13.7 %	52.4 %	8×	12×
380	2.4 %	49.7 %	4×	12×
460	−1.2 %	47.5 %	0×	12×
600	−5.3 %	46.5 %	0×	12×

The headroom never drops below 46 % at any K_0 , and the feasibility axis never binds at any K_0 — the safe disturbance scale is pinned at the top of the probe range throughout. The whole monotone-collapse story of §8.1 is a property of a tight torque budget, and this pose does not have one.

The fix comparison collapses with it. All seven candidates of §8.3 now score **identically at the ceiling**: both guarantees hold to 16×, no tank deficit, and α_{nom} **never fires for any of them**. Decision 0’s fix, which quadrupled the safe range at `phri2`’s pose, is entirely inert here.

That is not an argument against adopting it — insurance that never pays out in the tested scenario is still the right thing to carry, and §8.2’s failure mode is real wherever the budget *is* tight. But it does mean **§8 is a diagnosis of a configuration, not of the architecture**, and the paper should present it that way. Two of §8’s conclusions also need narrowing: the “passivity peaks at intermediate K_0 ” shape inverts at this pose (§10.2), and `anisotropic` stops being harmful — 47.9 % headroom against certified’s 45.7 %, and the same 16× safe range, so at this pose it is merely useless rather than costly.

8.5 Merged Lemma 1’: what the fix actually guarantees, and what it costs

§10.6 exposed a gap between the lemma as stated and the controller as shipped. Across 20 resampled runs at §10.5’s cell, Lemma 1’s **precondition** held in 10 and its **conclusion** held in 20. Both cannot be right for a conditional guarantee: one whose condition fails half the time should fail half the time. What closes the gap is α_{nom} , which is not in the lemma at all.

`pir_lemma_check.py` states the repaired lemma and checks every clause on 4000 adversarially sampled ticks ($|e|$ to 0.35 m, $|v|$ to 1.2 m/s, $|F_r|$ to 600 N — far outside anything the QP emits).

Merged Lemma 1’ (nominal authorization, monotone).

Hypotheses. (P1) $|\tau_{\text{base},\ell}| \leq \bar{\tau}$ at every tick. (P2) $E_0 \geq E_{\min}$. (P3) α_{nom} is monotone non-increasing.

Conclusions. (C1) $|\tau_\ell| \leq \bar{\tau}$ for all ℓ . (C2) $E_\ell \geq E_{\min}$ for all ℓ . (C3) until the first tick at which Lemma 1’s original precondition fails, $\alpha_{\text{nom}} = 1$ and the controller is bit-identical to the one without it.

(P1) is the whole precondition, and it is much weaker than Lemma 1’s. Lemma 1 required $|\tau_{\text{base}} + J_v^\top F_{\text{nom}}| \leq \bar{\tau}$, which depends on K_0 , D_0 , e and v — every design choice and the entire trajectory. (P1) depends on none of them: it is a property of the pose, and it uses 0.636 of joint 4’s cap at `phri2`’s pose and 0.433 at the recommended one. That is the real content of decision 0.

clause	result over 4000 adversarial ticks
(C1) torque envelope	0 violations , worst $\ \tau\ /\bar{\tau} = 1.000000$
(C2) tank floor	0 violations , worst $E = 0.020000 \text{ J} = E_{\min}$
(C3) inertness before the latch	0 violations over 1326 applicable ticks

(P3) is load-bearing, not cosmetic. The ledger debits $\frac{1}{2}(\alpha_{\text{nom}} - \alpha_{\text{prev}})e^\top K_0 e$ when the nominal *re-stiffens*, and nothing bounds that debit against the available energy. Monotone, the term is identically zero and (C2) goes through. Drop (P3) and the same 4000 ticks give **3997 floor violations, worst $E = -12\,700 \text{ J}$** . The monotone rule is not a tuning choice; it is what makes the energy half provable.

(C5) — α_{nom} is a latch, and the first draft of (C3) was wrong

(P3) makes α_{nom} monotone over the *whole run*, so once it fires it never recovers. The first version of this section stated (C3) without the “until the first tick”, and the checker falsified it on **1326 of 4000 ticks** — correctly. After the latch fires the controller is no longer inert even at ticks where Lemma 1’s precondition holds again: all 1326 differ.

That makes the per-contact reset §11 lists as owed a **correctness requirement rather than a nicety**: without it, one transient permanently softens the nominal for the rest of the run, and (C3) is empty beyond the first firing.

(C4) — the cost, and it re-opens decision 0

Where $\alpha_{\text{nom}} < 1$ the rendered nominal is $\alpha_{\text{nom}}K_0$, so the storage is $H_0 = \frac{1}{2}v^\top \Lambda v + \frac{1}{2}\alpha_{\text{nom}}e^\top K_0 e$ and the $\alpha \rightarrow 0$ fallback displacement scales as $1/\alpha_{\text{nom}}$. Diagnostic 3’s guarantee — that the fallback holds the push inside the box — is **not** preserved:

α_{nom}	effective K_0	fallback $\ e_z\ $	inside the 60 mm box
1.00	380 N/m	55.8 mm	yes
0.75	285 N/m	76.4 mm	no
0.50	190 N/m	117.0 mm	no

And the closed-loop exchange rate, against the ablation with α_{nom} disabled, is worse than it looks stated abstractly:

disturbance	pir : max $\ \tau\ /\bar{\tau}$	pir excursion	ablation: max $\ \tau\ /\bar{\tau}$	ablation excursion
1×	1.0000	100 %	1.0000	100 %
8×	1.0000	180 %	1.0074	109 %
12×	1.0000	172 %	1.0629	132 %

At 8×

disturbance, α_{nom} buys a **0.74 %** torque overrun by spending **71 percentage points** of workspace excursion — 108 mm instead of 65 mm against a 60 mm box. And the overrun it prevents is over the *derated* cap: $1.0074 \times 31.5 = 31.7 \text{ N}\cdot\text{m}$, which is **36 % of the FR3’s 87 N·m hardware limit**.

So the honest one-line reading of decision 0 is:

α_{nom} converts a torque-envelope violation into a workspace-bound violation. It preserves (C1) and (C2) at the cost of (C4), and (P3) buys (C2) at the cost of (C5).

Whether that is the right trade depends on what the derated envelope means, and this document has never said. If $\rho\tau_{\text{max}}$ is a thermal or duty-cycle budget, a 0.7 % transient overrun is nothing and 43 mm of extra excursion is a poor price. If it is a safety-certified limit, the trade is right. Decision 0 was adopted on the strength of (C1) and (C2) before (C4) was measured; §12 re-opens it with the exchange rate attached.

8.6 What this does and does not settle

Findings 1–3 are one problem with a known knob and a priced trade. Finding 4 is solved. What remains open is the same thing §7.7 pointed at: even with α_{nom} , the guarantee is “torque legal provided τ_{base} fits”, and nothing here defends *that*. On this FR3 pose τ_{base} uses 0.636 of joint 4’s cap with no control authority applied at all, so the margin is real but finite, and it is a pose property — which is why re-running the gate at a pose that does not load joint 4 remains the highest-value untried experiment (§10).

9. The pose study: most of this was the scenario

Every number above comes from `phri2`’s nominal pose under its $-z$ push. §8.4 flagged that the residual condition — τ_{base} alone fitting the envelope — is a *pose property*, and that re-running the gate elsewhere was the highest-value untried experiment. It was.

9.1 Screening poses, and two traps

`pir_pose_study.py` screens 1863 in-limit poses (varying q_2, q_4, q_6 ; 1169 of them usable, i.e. end effector above the base plane and out in front) across three push directions, on two cheap quantities: the gravity floor $\max \|g(q)\|/\bar{\tau}$, and the fallback-anchor proxy $g(q) - J_v^\top F_h$, which is what the anchor tends to as K_0 grows (§6.2).

Trap 1: the proxy rewards poses that cannot render impedance. Its first pick, $(q_2, q_4, q_6) = (-0.20, -0.90, 2.60)$, clears $\rho = 0.28$ with an anchor ratio of 0.685 — and is near-singular. Its task-space inertia along the push axis is **47.2 kg** against the neutral pose’s 4.56 kg, and $\sigma_{\min}(J_v)$ falls from 0.256 to 0.080. The arm is *braced*, not better: at 47 kg of apparent inertia the realization layer cannot render a 2 kg desired impedance in the push direction at all. A pose is therefore only a candidate if it also keeps the push direction controllable — Λ_{axis} within $2\times$ of neutral and $\sigma_{\min}$ at least $0.7\times$ — and both thresholds are stored per pose so a different line can be drawn without re-running.

Trap 2: the static proxy is optimistic, sometimes by a factor of three. The gate uses $\max_t \|\tau_{\text{base},t}\|_\infty/\bar{\tau}$ *along the trajectory*, where τ_{base} also carries Coriolis terms and the null-space and orientation torques that become nonzero the moment q leaves q_{null} . One shortlisted pose scored **0.470 statically and 1.287 along the trajectory**, with the binding joint moving from 4 to 5. That is the other half of the lesson: relieving joint 4 can simply move the bottleneck to joints 5–7, whose $\rho = 0.28$ caps are only 3.36 N·m. So the screen shortlists and a replay decides.

Decision 2: fallback-anchor proxy over pose and push direction. $q_6 = 1.571$; star = phri2's pose; black line = 1; blank = rejected as uncontrollable in the push direction (braced by kinematics, not better)

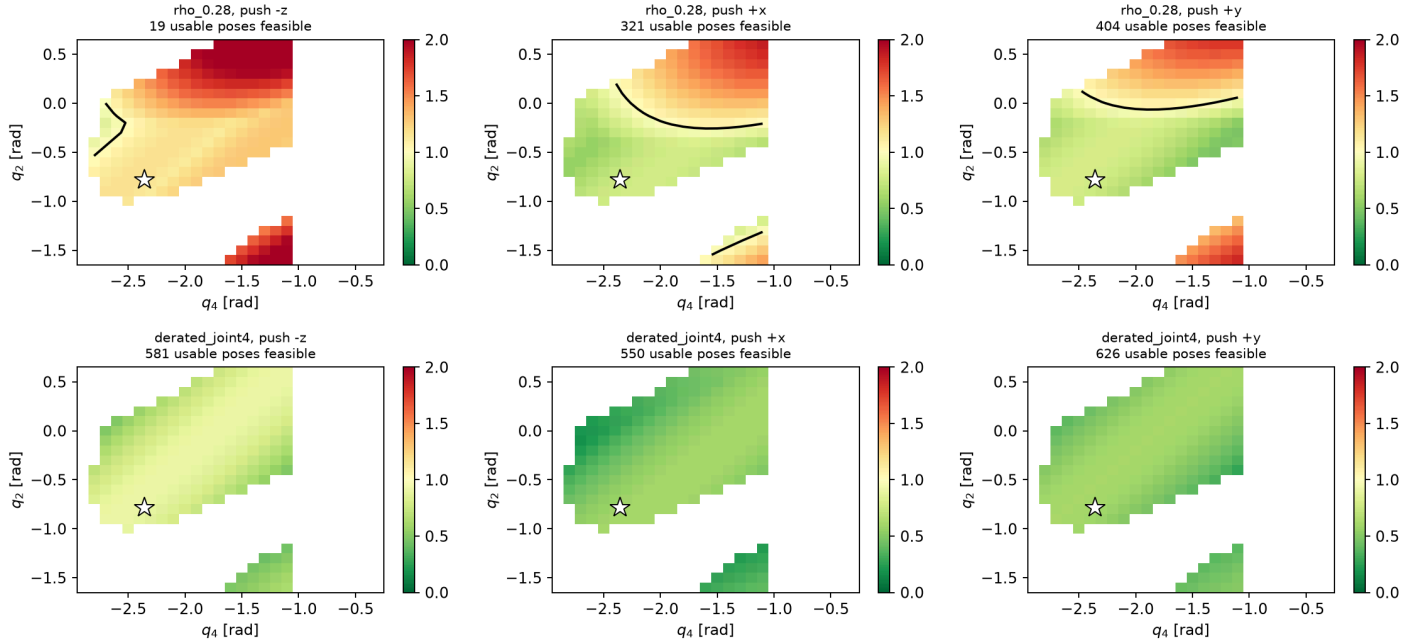


Figure 6 — the pose screen. Anchor proxy over (q_2, q_4) at $q_6 = 1.571$, per envelope and push direction. The star is phri2's own pose. Blank cells are rejected by the controllability guard — poses that resist the push through kinematic structure rather than control, which the unguarded screen preferred. Green is not yet a verdict: §9.1's second trap means these still have to survive an along-trajectory replay.

9.2 $\rho = 0.28$ is reachable — but not by enough

§5.1's "structurally infeasible" was too strong, and is corrected here. It is infeasible at phri2's pose and push direction. Elsewhere it is not: at $(q_2, q_4, q_6) = (-0.10, -2.70, 1.571)$ under an $+x$ push, phri2's own realized torque peaks at **0.952** of the $\rho = 0.28$ cap — its published controller runs inside impedance_residual's envelope there. The gain-independence argument of §6.2 stands; what was wrong was calling a scenario property a structural one.

It still does not help the merge. The best along-trajectory floor found anywhere in the shortlist is **0.870**, leaving $\leq 13\%$ of the cap for $J_v^T F_{\text{nom}}$ — against the 36.4 % the neutral-pose derated envelope offers, which supported only a 33-cell sliver. A passive nominal does not fit in 13 %.

9.3 The derated envelope at a better pose: the actual result

Re-running the full gate at $(q_2, q_4, q_6) = (-1.30, -1.30, 1.571)$, keeping phri2's own $-z$ push so only one thing changes:

	phri2 's pose	this pose
$\tau_{\text{base floor}}$	0.636	0.433
phri2 's own realized torque	1.005	0.508
cells passing row (1)	577 / 900	900 / 900
cells passing row (1b)	702 / 900	900 / 900
common feasible region	33 cells	481 cells
most robust cell	$K_0=380, D_0=29.1$	$K_0=520, D_0=56.1$
worst normalized use there	0.977	0.541

And in closed loop, at each pose's own recommended operating point, sweeping the disturbance:

disturbance	headroom (phri2 pose)	headroom (this pose)	residual share	$\max \ \tau\ /\bar{\tau}$	$\max \ e_z\ $
1×	2.4 %	45.7 %	17.1 % → 54.0 %	1.0000 → 0.532	60.2 → 51.5 mm
8×	−16.8 %	43.2 %	72.4 % → 86.7 %	1.0000 → 0.533	108.0 → 58.9 mm
16×	−55.5 %	35.4 %	85.0 % → 90.7 %	1.0000 → 0.619	164.3 → 59.2 mm

Three things follow.

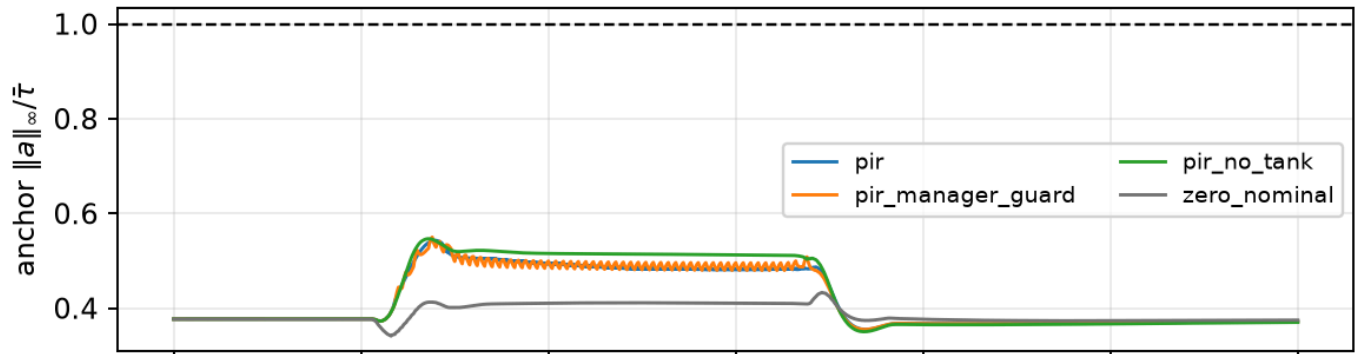
Findings 1 and 3 are largely gone. The residual is 54 % of the command instead of 17 %, with 45.7 % of joint 4's cap behind it instead of 2.4 %.

Finding 4 stops being stressed at all. At 16× disturbance the anchor still has 35 % headroom and the applied torque peaks at 0.62 of the cap — the controller is not saturating, so Lemma 1's precondition is never in question. Decision 0's α_{nom} remains the right insurance (the negative headroom entries in the phri2 -pose column are exactly it absorbing an overrun that would otherwise have been a violation), but at this pose it is not called on.

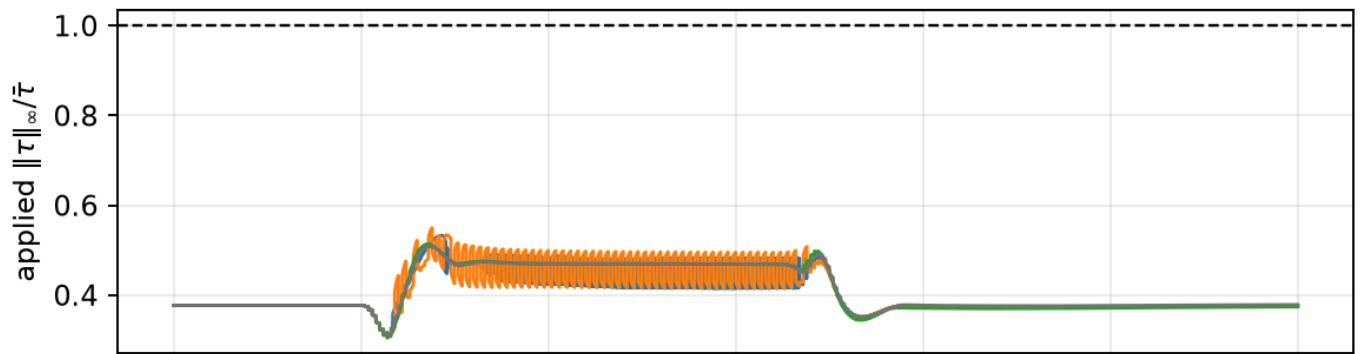
It is better on the workspace too, not traded against it. 59 mm at 16× disturbance against 164 mm — inside the 0.06 m box where the certified point is 2.7× outside it. This is the opposite of the K_0 trade §8.3 priced, which bought headroom *with* containment. **Decision 1 may therefore be moot:** the pose change delivers what softening K_0 was being considered for, and improves the thing softening K_0 would have cost.

PIR merged controller, scenario 'push', envelope 'derated_joint4'

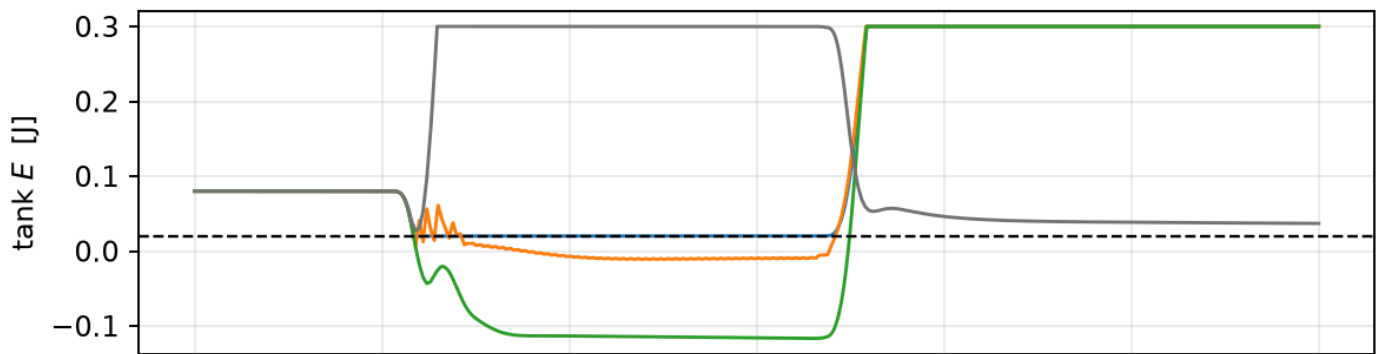
Merged Lemma 1's precondition (row 1), measured in closed loop



Merged Lemma 1's conclusion



energy tank vs its floor $E_{\min} = 0.02$ J



workspace excursion on the push axis vs the 0.06 m bound

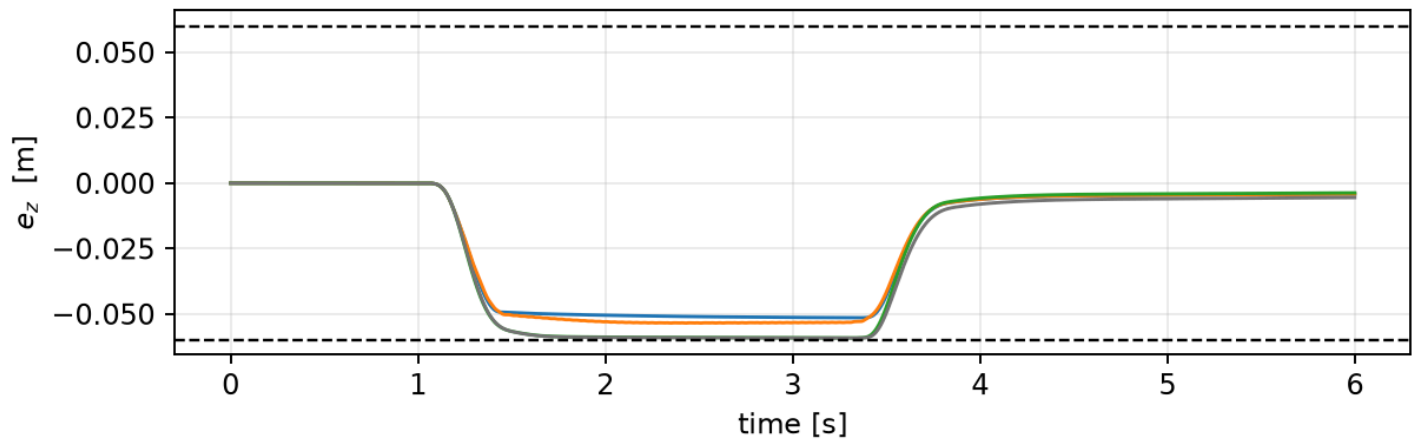


Figure 7 — the merged controller at the recommended pose, on `phri2` 's plain 20 N push. The third panel is the one to read: `pir` (blue) rides *exactly* on $E_{\min}$ for the whole push — it is under the dashed line, which is the guarantee made visible — while `pir_manager_guard` (orange) sits below the floor for roughly two seconds and `pir_no_tank` (green) plunges to -0.117 J. `zero_nominal` (grey) saturates at $E_{\max}$ because with no nominal its residual *is* the impedance and is purely dissipative. Compare the top two panels with Figure 2: everything now runs at 0.4-0.55 of the envelope rather than pinned against it.

9.4 Tasks 2 and 3 re-run at the recommended pose

Everything in §7 and §8 was measured at `phri2` 's pose. The merged paper should report the pose it recommends, so both tasks were re-run at $(q_2, q_4, q_6) = (-1.30, -1.30, 1.571)$ with that pose's own operating point, $K_0 = 520$ N/m, $D_0 = 56.1$ N·s/m.

A bug surfaced first, and the four-term residual is what caught it. The realization QP and the 1 kHz servo each build τ_{base} , and the QP was centring the null-space posture spring on `Q_NEUTRAL` while the servo centred it on the actual pose — so the QP planned against a τ_{base} and a d_{known} the servo never applied. At `Q_NEUTRAL` the two coincide and the bug is invisible; at the new pose the closure went from 4×10^{-16} to 8×10^{-5} . Fixed, closure back to 1.3×10^{-15} , and a regression test now pins it at both poses. This is the second time the closure identity has caught something no other diagnostic would have.

Task 2 — the merged controller. On `phri2` 's plain 20 N push:

	<code>phri2</code> pose	recommended pose
row (1), replay → closed loop	0.9771 → 0.9762	0.5411 → 0.5432
anchor headroom	2.4 %	45.7 %
residual share of command	15.6 %	53.5 %
$\max \ \tau\ /\bar{\tau}$	1.0000	0.5330
authorization active	0.6 % of ticks	29.3 %
$\min \alpha_E$	1.0000	0.0006
r_{auth} RMS	0.0000	1.387 m/s ²
max excursion	60.4 mm	51.4 mm

The passivity axis is now exercised by `phri2` 's own benchmark. §7.4 had to borrow `impedance_residual` 's oscillatory disturbance to make the tank do anything, and concluded that neither source benchmark stresses both axes. At the recommended pose that is no longer true: on the plain push, authorization fires on 29 % of ticks, α_E reaches 0.0006, and `pir_manager_guard` breaches the floor for **2 039 ticks** while `pir_no_tank` reaches -0.117 J. One scenario, both axes, no borrowed disturbance — which is what the merged paper needs.

Task 3 — the authorization-vs-tracking trade, and a correction. The E_0 sweep at the recommended pose:

	<code>phri2</code> pose	recommended pose
E_0 trade, RMS spread	0.03 %	7.66 %
authorization active over the sweep	0.0-1.9 %	30.1-33.9 %
<code>pir</code> floor held	12 / 12	12 / 12
<code>pir_manager_guard</code> floor held	4 / 12	1 / 12 (21 138 ticks)
<code>pir_no_tank</code> floor held	4 / 12	1 / 12 (25 525 ticks)
tank's cost vs no tank at $E_0 = 0.08$	+0.0 %	+38.4 % RMS

So §7.5’s “flatter, not steeper” was a pose artifact and is corrected: the synthesis note’s prediction was right. Give the residual room and the passivity axis costs what the note said it would — 7.7 % across the E_0 sweep, and 38 % against running with no tank at all. It is no longer cheap-because-weak; it is doing real work and charging real money for it. The fast-vs-manager-rate result also gets stronger, holding across the *entire* E_0 range rather than only its bottom end.

PIR: authorization vs tracking with both tightenings active ($K_0 = 520.0$ N/m, $D_0 = 56.13$ N·s/m, envelope derated_joint4)

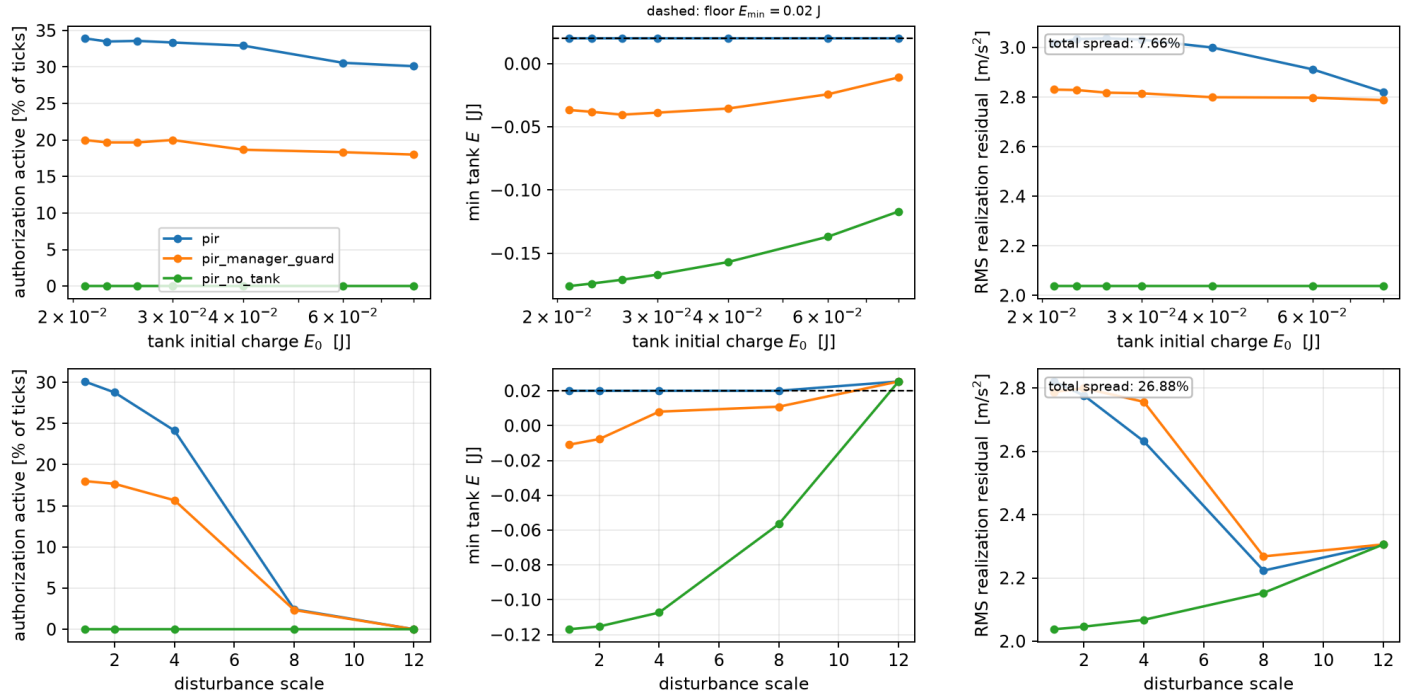


Figure 8 — authorization vs tracking at the recommended pose. Against Figure 3, the middle column now shows `pir` holding $E_{\min}$ across the whole sweep while both baselines breach everywhere, and the right column carries a 7.7 % spread instead of 0.03 % — a real trade rather than a flat line.

One counter-intuitive detail, stated because it is not obvious: authorization becomes *less* active as the disturbance grows (30.1 % → 0.0 % at 12×), the opposite of the `phr12` pose. The tank harvests $\alpha_{\text{nom}} v^T D_0 v$, which grows quadratically with the disturbance-driven velocity, while the residual’s debit $F_r^T v$ grows linearly — so a large disturbance fills the tank faster than it drains it. The stressing case here is the *nominal* scenario, not the extreme one.

9.5 What this does not settle

The pose was chosen by a screen tuned on one scenario, and only q_2, q_4, q_6 were varied. Nothing here says it is optimal, or that it is a pose a real task would want the robot in — it is a better *test* pose, and whether the application permits it is a question this study cannot answer. The two guard thresholds are judgement calls. And the whole comparison is still one seed of one disturbance profile on one arm in simulation.

10. The tension between the two axes

PIR’s name rests on realizability having *two* axes — feasibility, reported by r_{con} , and passivity, reported by r_{auth} — and the synthesis note is explicit that without the second axis the whole thing is a rebrand of constrained-MPC impedance. So the question that decides the framing is not whether each axis works. It is whether there is an operating point where **both are load-bearing at once and the controller is well behaved**.

On the evidence of the runs collected up to §10.4, there is not; §10.5 then scans pose and K_0 jointly and finds exactly one candidate, whose status depends on how the workspace box is read. `pir_axis_tension.py` aggregates every closed-loop run and sweep point in `results/` and classifies each by which authorization actually fired.

PIR's two axes are anti-correlated: each binds where the other does not

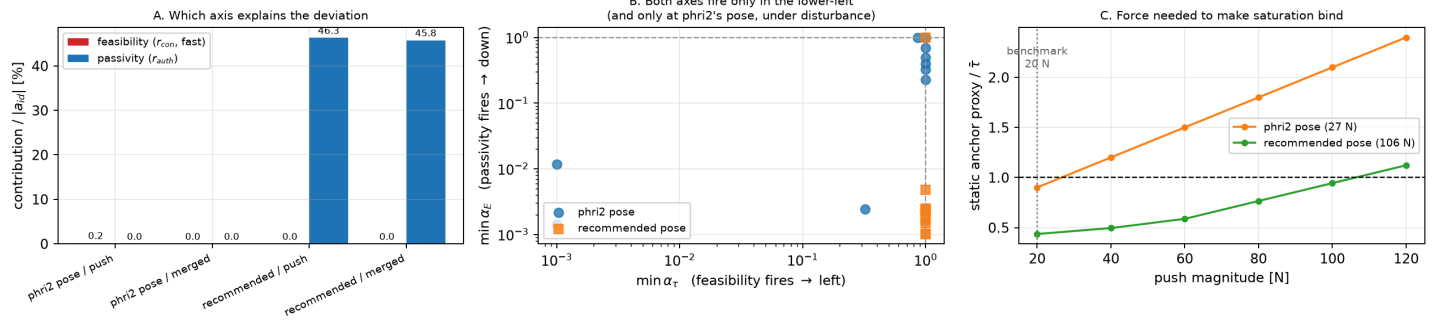


Figure 9 — each axis binds where the other does not.

10.1 Each axis is inert where the other is strong

Each axis's contribution is reported against $|a_{id}|$, the behaviour the controller is trying to render — the same denominator §10.5 uses, so the two are comparable. (Normalising by the *net* realization residual would not be a share of anything: the four terms sum to the net but oppose one another, so a single term can exceed it.)

run	$\min \alpha_\tau$	$\min \alpha_E$	$r_{\text{con}} / \ a_{id}\ $	$r_{\text{auth}} / \ a_{id}\ $	headroom
phri2 pose / push	0.8504	1.0000	0.2 %	0.0 %	2.4 %
phri2 pose / merged	1.0000	1.0000	0.0 %	0.0 %	2.4 %
recommended / push	1.0000	0.0006	0.0 %	46.3 %	45.7 %
recommended / merged	1.0000	0.0010	0.0 %	45.8 %	45.7 %

At the recommended pose the passivity axis removes **46 %** of the intended behaviour and the feasibility axis contributes **nothing** — α_τ stays at 1.0 across all 12 sweep points there, so the fast torque projection never once intervenes. At `phri2`'s pose the situation inverts: α_τ fires and r_{auth} is *exactly* zero.

Across all 24 sweep points, **4 have both axes firing** — and all four are at `phri2`'s pose under 4–12× disturbance, which is precisely the regime §9.3 showed to be badly behaved (negative headroom, 103–108 mm excursion against a 60 mm box). The one well-conditioned dual-axis point, $E_0 = 0.026$ at nominal disturbance, has $\alpha_\tau = 0.9999$ — the feasibility axis is technically firing and doing nothing.

10.2 Why — and one place where I over-generalised again

Both authorizations act on the **same object**, the residual F_r , through the **same** torque budget. When that budget is tight they compete: a nominal stiff enough for saturation to bind squeezes the residual (17 % of the command at $K_0 = 380$, `phri2` pose), and a small residual does little port work, so the tank never drains.

The K_0 sweep at `phri2`'s pose shows exactly that:

K_0	headroom	residual share	feasibility: safe disturbance	passivity: E_0 trade
100	26.4 %	72.6 %	12×	0.0068
200	17.9 %	46.4 %	8×	0.0125
250	13.7 %	34.1 %	8×	0.0358 ← strongest here
380	2.4 %	17.1 %	4×	0.0020
600	−5.3 %	48.4 %	0×	0.0010

I first wrote that up as a structural law — passivity peaks at intermediate K_0 , feasibility binds monotonically — and the re-run at the recommended pose (§8.4) contradicts it, which is the same over-generalisation §11.0 catalogues, committed while writing the section that catalogues it. At the recommended pose the headroom never falls below 46 % at *any* K_0 , the feasibility axis never binds at any K_0 , and the E_0 trade *rises* with K_0 (0.021 up to $K_0 = 380$, then 0.32 at 460) instead of peaking in the middle.

What survives is narrower and still sufficient:

- **The competition is real only when the torque budget is tight.** With ample headroom a stiff nominal does *not* squeeze the residual — at the recommended pose $K_0 = 460$ still leaves the residual 50 % of the command — so the two axes decouple. But they decouple into a regime where feasibility never binds at all, which is not a dual-axis regime either.
- **The axis-firing evidence is pose-robust and is what §10.1 rests on**, since it is measured directly rather than inferred: α_r never once fires across 12 sweep points at the recommended pose, and r_{auth} is exactly zero across the headline runs at `phri2`’s.
- **Only a large disturbance loads both at once**, because it raises velocity (draining the tank) and the anchor (loading the envelope) together. That is why all four dual-axis points are high-disturbance ones.

So the conclusion of §10.1 stands — no well-behaved dual-axis operating point exists in the data — but the tidy mechanism I reached for does not, and §10.5’s joint scan is the way to find out which story is right.

10.3 The recommended pose optimises away PIR’s own feasibility half

Panel C quantifies it. Taking the static anchor proxy of §9.1 — which is *optimistic* about feasibility, so these are lower bounds:

pose	τ_{base} floor	push needed for the anchor to reach $\bar{\tau}$
<code>phri2</code>	0.602	27 N
recommended	0.377	106 N

The benchmark push is 20 N, which is already a firm two-handed shove. At the recommended pose you would need roughly **five times** that before saturation became the binding constraint — outside the range a human interaction produces.

So the pose that makes the passive-nominal split work is a pose where the saturation story PIR inherits from `phri2` does not arise. That is an uncomfortable sentence and it belongs in the paper, not in a footnote.

10.4 What this means for the framing

It does **not** mean the second axis is unearned. It means the claim has to be stated as what the evidence supports, which is a *stronger* claim than “both axes always bind”:

*The value of a two-axis certificate is not that both axes bind together. It is that **you cannot tell in advance which one will bind**. This study changes one FR3 joint angle — a change a practitioner would consider innocuous, and which improves every behavioural metric — and the binding axis flips from feasibility to passivity. A single-axis certificate is silent after that flip: at*

the recommended pose `phri2`'s r_{con} reports zero while half the realization deviation is energy authorization, and at `phri2`'s pose `impedance_residual`'s tank reports zero while the torque projection is the only thing intervening.

That is what the four-term residual buys, and it is demonstrable on the data already collected. The honest presentation is two case studies — a saturation case at `phri2`'s pose and a passivity case at the recommended one — with §10.2 as the explanation for why one experiment cannot be both.

10.5 The joint scan: one candidate, and it hangs on a judgement call

Pose and K_0 had only ever been swept separately, so the two sweeps could have been cutting across a diagonal ridge neither resolved. `pir_joint_scan.py` scans them jointly: a pose family interpolating in joint space between `phri2`'s pose ($\lambda = 0$) and the recommended one ($\lambda = 1$), crossed with $K_0 \in [150, 600]$ and disturbance $\in \{1\times, 4\times\}$ — 96 cells, with D_0 set from a fixed damping ratio against each pose's own task-space inertia so K_0 is the only gain varying.

“Load-bearing” is deliberately stricter than “fired once”: an axis counts only if its residual term removes at least 2 % of the intended behaviour $|a_{id}|$. (Reported as a *ratio*, not a share — under a $4\times$ disturbance the QP's residual fights something much larger than a_{id} , so the ratio legitimately exceeds 1. An earlier version of this script normalised by the *net* realization residual, which is not a share at all: the four terms sum to the net but oppose one another, and it produced “shares” of 468 %.)

Result: 41 cells are well behaved, 5 are dual-axis, and the intersection is empty at the headline box allowance. In every one of the 41 well-behaved cells r_{con} is *identically zero* — even in the three with under 5 % anchor headroom, where α_τ still never fires because the tight budget has squeezed the residual small enough to fit in what is left. That is §10.2's surviving mechanism, confirmed on a grid rather than inferred from two points.

All 5 dual-axis cells sit at $\lambda = 0$ — `phri2`'s own pose — under $4\times$ disturbance. And they fail “well behaved” for a reason worth stating precisely: **not because the certificate breaks**. Lemma 1's precondition holds, its conclusion holds, the tank floor holds, and α_{nom} never fires in any of them. They fail on **workspace containment** — 69.5 to 134.7 mm against the 0.06 m box.

Joint pose x K_0 scan: is there a well-behaved dual-axis operating point?

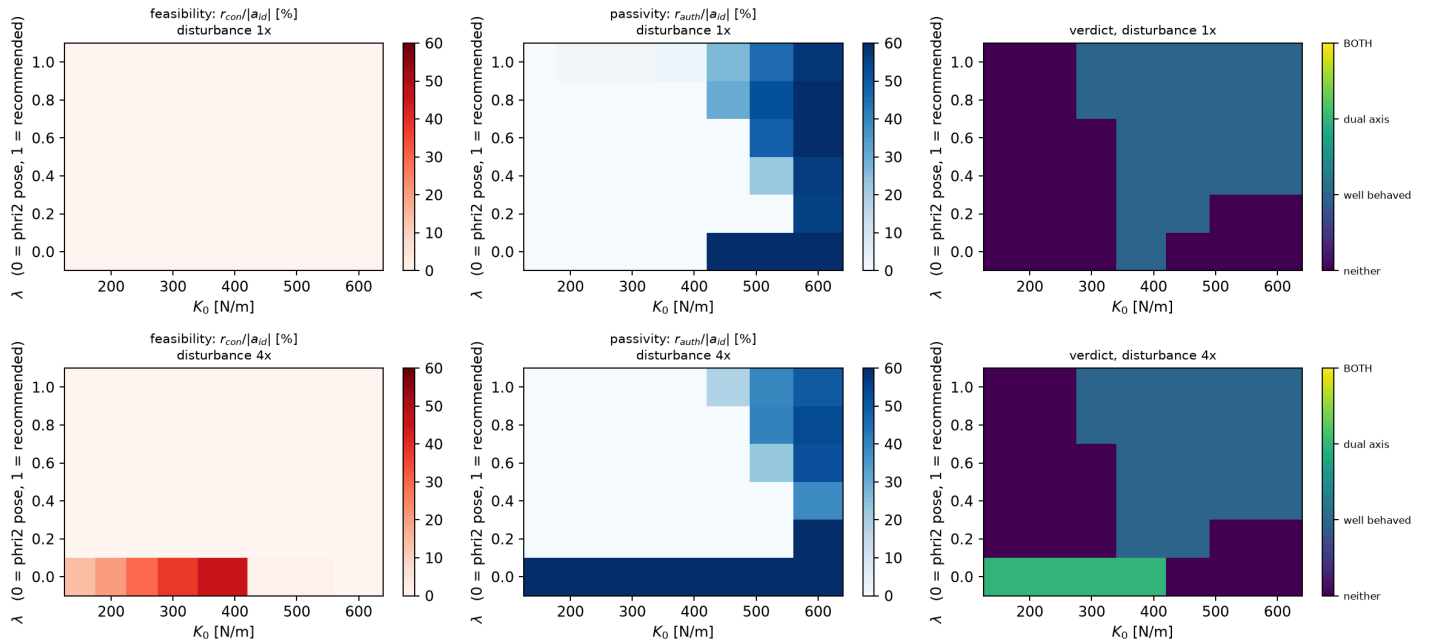


Figure 10 — the joint pose x K_0 scan. Left column: the feasibility axis, which has colour only along the bottom edge ($\lambda = 0$, `phri2`'s pose). Middle: the passivity axis, concentrated at high K_0 and high λ . The two never light up together anywhere the verdict panel (right) calls well behaved — no cell is marked “BOTH”. The one candidate of the next subsection is at $\lambda = 0$, $K_0 = 380$, and it is classified here as dual-axis but not well behaved, because the 10 % box allowance this figure uses excludes it.

The candidate

That makes the verdict turn on how much overshoot of a **slack-relaxed** box counts as a violation, which is a judgement call the scan should not make silently:

box allowance	dual-axis <i>and</i> well behaved
1.00× - 1.15×	0
1.20× - 1.30×	1 — $\lambda = 0$, $K_0 = 380$, 4× disturbance

At a 20 % allowance exactly one cell qualifies, and it is a striking one: `phri2` 's **own pose at the originally certified operating point**, $(K_0, D_0) = (380, 29.07)$ — the point this whole study started from — under 4× the source disturbance:

$r_{\text{con}} / \ a_{id}\ $	45.6 %
$r_{\text{auth}} / \ a_{id}\ $	95.3 %
$\min \alpha_\tau / \min \alpha_E$	0.320 / 0.0024
α_{nom}	1.0 (decision 0's fix not needed)
anchor headroom	+1.1 %
Lemma 1 precondition / conclusion	hold / hold
tank floor	holds
fallback $\ e_z\ $ (diagnostic 3)	55.8 mm ✓
excursion	69.5 mm , 16 % over the box

Both axes carrying 46 % and 95 % of the intended behaviour, both α 's deep into intervention, and every certificate intact. If `phri2` 's box is read as what it is in its own QP — slack-relaxed, a preference under model mismatch rather than an actuator limit — this is the dual-axis demonstration the paper needs, and §10.1's "no such point" becomes "exactly one, and it is the obvious one".

This promotes a question that was already on the owed list to a decisive one. §12 has carried "resolve whether the 0.06 m bound is hard or slack-relaxed" since the Task 1 gate, where it decided a 33-cell versus 6-cell region. It now also decides whether PIR's central claim has a supporting experiment. That convergence is not a coincidence — both are asking the same thing, whether the workspace box is a specification or a preference.

What it does not buy

One cell, at 4× the source disturbance, 16 % outside the nominal box, at one seed on one arm. It is an existence proof, not an operating recommendation — and note that it is *not* the pose §9 recommends. The honest reading is that PIR's two axes can be made simultaneously load-bearing, but only by running the arm harder than either source paper does, at the pose with the tight torque budget, and accepting an excursion the recommended pose avoids.

10.6 Both formulations, tested across seeds

§12's decision 3 offered two ways to state the claim, and both rested on a single disturbance seed — (b) on a single cell at a single seed. `pir_robustness.py` re-runs both across 10 seeds × 2 disturbance profiles (with and without `rejectable_force` 's between-manager-tick pulse, the component `impedance_residual` added specifically to defeat a slow guard). 60 runs.

Decision 3: robustness of formulation (a) the flip, and (b) the dual-axis candidate

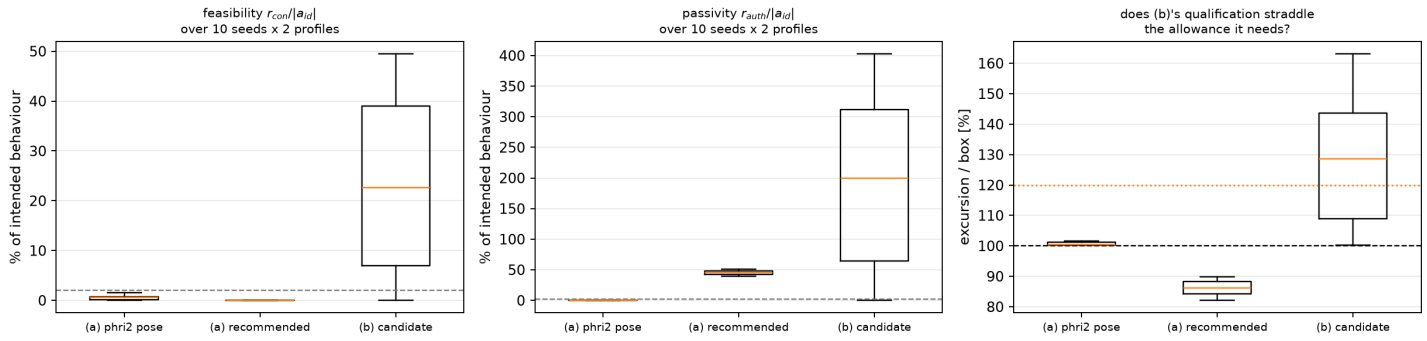


Figure 11 — decision 3 under resampling.

(a) The flip survives, 20/20 — with one correction to how it is stated

	phri2 pose	recommended pose
$r_{auth}/ a_{id} $	exactly 0 in 20/20	39.8 - 50.6 % in 20/20
$r_{con}/ a_{id} $	0.00 - 1.51 %	exactly 0 in 20/20
classification	neither axis load-bearing, 20/20	passivity-only, 20/20

The assignment never swaps. But the strict load-bearing threshold exposes something §10.4's wording glosses: at $1\times$ disturbance the feasibility axis **never reaches 2 % of the intended behaviour at either pose** — it fires (α_r ranges 0.77-1.00 at phri2 's pose) but what it removes is under 1.5 %. So the robust statement is not “the load moves from one axis to the other”. It is:

*The passivity axis goes from **exactly zero** to **40-51 %** of the intended behaviour, and the feasibility axis from small-but-nonzero to **exactly zero**, on a joint-angle change that improves every behavioural metric. What flips is **which axis is capable of carrying load at all**.*

That is still the claim §10.4 wants and it is now 20/20 rather than 1/1. It is weaker than “the axes exchange the load”, and the paper should say the weaker thing.

(b) The exhibit does not survive as stated

	result
dual-axis and every certificate intact	5 / 20
Lemma 1 precondition holds	10 / 20
Lemma 1 conclusion holds	20 / 20
tank floor holds	20 / 20
excursion, all runs	100 - 163 % of the box
excursion, runs that qualify	111 - 120 %

§10.5's cell was seed 0 with the pulse — one of the lucky draws. Resampled, the candidate is dual-axis in 15 of 20 runs but keeps **all** its certificates in only 5, and the failure is always the same one: **Lemma 1's precondition**, which holds in exactly half the runs. (The conclusion and the tank floor never fail — α_{nom} and the fast re-authorization do their jobs every time. That is a real result in itself, and it is decision 0's fix earning its place.)

The pattern in the failures is §10’s anti-correlation appearing *within a single cell*: the runs with the largest r_{auth} (122–403 %) and the largest excursions (137–163 %) are exactly the runs where the precondition breaks. Loading both axes harder does not get you a better exhibit; it gets you a broken anchor.

So (b) is not a one-off fluke — it recurs at three distinct seeds — but it is a minority outcome (25 %) that always sits outside the nominal box (111–120 % even when it qualifies). It is not something to build a paper’s central claim on. §12’s decision 3 resolves to **(a), with (b) demoted to a remark**: that a dual-axis regime exists at all is worth one sentence and a pointer to the data; it is not the exhibit.

11. What this does not show

11.0 A pattern in what turned out to be wrong

Nine conclusions in this document were later overturned or narrowed by a subsequent experiment, and the pattern is worth stating because it bears on how much anything here should be trusted:

conclusion	fate
$\rho = 0.28$ is <i>structurally</i> infeasible (§5.1)	✗ pose-specific (§9.2)
the E_0 trade is <i>flatter</i> , not steeper (§7.5)	✗ pose artifact (§9.4)
neither source benchmark stresses both axes (§7.4)	✗ pose artifact (§9.4)
the nominal dominates the command, 84 % (§7.6)	✗ pose artifact, → 46 % (§9.4)
the split makes a hard guarantee conditional (§7.7)	✓ true, but fixable (§8.2) and inert at the good pose (§8.4)
anisotropic gains do not help <i>and slightly hurt</i> (§8.3)	~ the “hurt” is pose-specific; useless but harmless at the good pose (§8.4)
passivity peaks at intermediate K_0 (§10.2, first draft)	✗ inverts at the good pose — committed <i>while writing this table</i>
α_{nom} is “inert while the anchor fits” (§8.2)	~ only until its latch first fires; §8.5’s (C5) falsified the unqualified form on 1326 of 4000 ticks
decision 0 “costs nothing measurable” (§12, as adopted)	✗ §8.5’s (C4): 71 points of excursion for 0.74 % of a derated cap

Five of nine were properties of **one FR3 configuration**, not of the architecture; two more (the last two rows) were claims about the *fix* that its own checker falsified — and in each case the erroneous generalisation was made from a carefully measured, internally consistent experiment. The measurements were right; the scope claimed for them was not. **Nothing in this line should be claimed without a pose sweep** — a rule I restated in §10.2 and then broke in the next paragraph, which is the most honest evidence available for how strong the pull toward the tidy generalisation is. The two bugs that the four-term residual’s closure check caught (§8.2, §9.4) suggest the same about any diagnostic that cannot be checked against an identity.

- ~~The merged controller is not implemented.~~ Discharged in §7.2: the replay estimate and the merged closed loop agree to 0.1 % on both rows, and a regression test now pins them together.
- ~~The anchor’s feasibility does not survive a larger scenario.~~ Fixed in §8.2 by `nominal_auth_mono`, which holds both guarantees to 16×. What is *not* fixed is the condition underneath it: everything still assumes τ_{base} alone fits the envelope, which is a property of the pose and is not defended anywhere.
- **The fix’s cost is not fully characterised.** `nominal_auth_mono` ratchets the floor’s stiffness down and never recovers it within a run. A 6 s benchmark does not show what that does over minutes of interaction, and the per-contact reset a deployed system would need is not implemented or tested.
- **The K_0 trade is priced on one disturbance profile.** §8.3’s workspace numbers at 12× come from `impedance_residual`’s own rejectable force scaled up, at one seed. That is a stress test, not a distribution.

- **The fallback leaves the workspace box in transient.** 62.2 mm peak against a 60 mm bound. `phri2` 's box is slack-relaxed rather than hard, so this is not a constraint violation in its formulation — but it means the $\alpha \rightarrow 0$ guarantee is “settles inside the box”, not “stays inside it”. If the bound is to be read as hard, the region is 6 cells, not 33, and the recommended cell is outside it.
- **One pose, one push direction, isotropic gains.** The whole gate is evaluated at `phri2` 's nominal hold pose under a $-z$ push, with $K_0 = kI$, $D_0 = dI$. Joint 4 is the binding joint in every single cell, and it is binding largely because of an 18.96 N·m bias torque at that specific configuration. An anisotropic (K_0, D_0) — stiffer along the push axis, softer elsewhere — is the obvious first lever if the region needs widening, and the note explicitly defers it. A different pose could move the verdict in either direction.
- **Recursive feasibility is untouched**, in both source papers and here. The combined torque-envelope + energy-floor + workspace-slack feasibility question is harder jointly than separately and cannot be self-certified.
- **The force-misclassification pillar is untouched.** The tank meters $F_r^\top v$; if F_h leaks into $\hat{d}$, or F_e is folded into the disturbance, the power sign is wrong and every certificate here stays green while the arm pushes a mislabelled wall. **No green cell in this document implies anything about that pillar.**
- **Double tightening is not measured.** `phri2` 's horizon-wide torque tightening and `impedance_residual` 's tank tightening compound in series. That curve is Task 3 and is expected to get steeper after the merge.
- Simulation only, translational only, affine memoryless behaviour class.

12. Decision gate

The synthesis note's §13.3 put the decision gate right after Task 1. The work ran well past that, because each result changed what the next question was; what follows is where it actually stops.

`common_region_nonempty == true` under `phri2` 's derated-joint-4 envelope, the merged controller is implemented, Merged Lemma 1 holds in closed loop and the four-term residual closes exactly. What is left is not a coding decision.

§8, §9 and §10 have each changed what is being decided. Finding 4 is fixed and adopted; findings 1–3 turned out to be largely the pose; and §10 has moved the open question from the design to the **framing**. Two decisions are settled and two are open.

Decision 0 — adopt `nominal_auth_mono` ? ADOPTED, but §8.5 re-opens it. It quadruples the range over which both guarantees survive and is provably inert until its latch first fires, and §8.4 showed it is inert at the recommended pose throughout — insurance rather than a load-bearing part. Two things found since it was adopted change the calculus at the operating points where it *does* fire:

- **The exchange rate (§8.5, C4).** At 8× disturbance it buys a 0.74 % overrun of the *derated* cap — 31.7 N·m against an 87 N·m hardware limit — by spending 71 percentage points of workspace excursion, 108 mm instead of 65 mm against a 60 mm box. Whether that is right depends on what $\rho \tau_{\max}$ *means*, which this document has never stated. **That is the question to answer before hardware**, and it is a question about the specification, not about the controller.
- **The latch (§8.5, C5).** α_{nom} never recovers within a run, so one transient permanently softens the nominal. The per-contact reset is a correctness requirement, not a nicety.

Decision 3 — how is the two-axis claim stated? SETTLED by §10.6: formulation (a), with (b) demoted to a remark. Both candidates were resampled across 10 seeds × 2 disturbance profiles:

- **(a) The flip — survives 20/20.** The passivity axis goes from exactly zero at `phri2` 's pose to 40–51 % of the intended behaviour at the recommended one, and the feasibility axis from small-but-nonzero to exactly zero, on a joint-angle change that improves every behavioural metric. One correction §10.6 forces: at 1× disturbance the feasibility axis never reaches the load-bearing threshold at *either* pose, so the claim is “**which axis can carry load at all flips**”, not “the axes exchange the load”. Say the weaker thing.
- **(b) The dual-axis exhibit — does not survive as stated.** §10.5's cell was a lucky seed. Resampled, it is dual-axis in 15/20 runs but keeps all its certificates in **5/20**, always failing on Lemma 1's precondition, and even when it qualifies it sits 111–120 % outside the box. It recurs at three distinct seeds so it is not noise, but a 25 % outcome is not a central claim's evidence.

This also **retires the box question as a framing blocker**. It still matters for the Task 1 region size, but the exhibit it would have licensed does not survive resampling, so the framing no longer waits on it.

Decision 1 — how much workspace containment under disturbance is the predictive layer’s authority worth?

Probably moot now. This was the real form of findings 1-3: $K_0 = K_d = 200$ buys $6.8\times$ the anchor headroom and $2.7\times$ the residual share, and pays with 160 mm of excursion at $12\times$ disturbance against 79 mm. §9.3 gets $19\times$ the headroom and $3.2\times$ the residual share from the pose instead, and *improves* the excursion to 59 mm. Unless the application pins the pose, take the pose and leave K_0 alone. Still worth a human’s eye, because “the application pins the pose” is exactly the kind of constraint this study cannot see.

Decision 2 — which envelope does the merged paper claim? SETTLED by §9: `phri2`’s derated-joint-4 envelope, at a pose that does not load joint 4. $\rho = 0.28$ is reachable (§9.2 corrects §5.1’s over-claim) but leaves at best 13 % of the cap for the nominal, which is not enough for the split. For the record, the three options as they stood before §9 ran:

1. **Write the merge against `phri2`’s derated-joint-4 envelope.** $(K_0, D_0) = (380, 29.1)$, region of 33 cells, margins of $2-7\%$. Honest, and the source benchmark actually runs there. Costs: the tank’s numbers from `impedance_residual` were measured under $\rho = 0.28$ and would have to be re-derived, and the margins are thin enough that §7’s first bullet could consume them.
2. **Keep $\rho = 0.28$ and abandon the passive-nominal split**, retreating to “dissipativity directly on the realized port, no nominal” — a different and harder design, and a separate planning decision. This looked *forced* when §5.1 read its result as structural; §9.2 withdrew that, so it was never forced — only expensive.
3. **Change the scenario** — a different nominal pose, a lower push magnitude, or a push direction that does not load joint 4 — and re-run this gate. Cheap to try (the scan is ~ 19 min on 4 cores) and it attacks the actual binding quantity, which is an 18.96 N·m bias torque at one configuration rather than anything about impedance.

Option 3 is what §9 ran, and it won: the pose relieves findings 1, 3 and 4 at once and improves containment rather than trading it. Option 1 is therefore the recommendation, *at the §9.3 pose rather than `phri2`’s*. Option 2 is not forced — the passive-nominal split is viable, so there is no reason to retreat to dissipativity-on-the-realized-port.

A tension that decision 3 has now dissolved rather than resolved: decision 2 recommends the §9.3 pose while §10.5’s dual-axis cell lives at `phri2`’s, so formulation (b) would have had the paper recommending one operating point and demonstrating its central claim at another. §10.6 demoted (b), so the question no longer arises — but it is recorded because it is the kind of thing that would have been easy to gloss.

Owed before any claim

- ☒ ~~Decision 0: adopt `nominal_auth_mono`~~ — adopted; it is the controller’s default, with `pir_no_nominal_auth` kept as the ablation.
- ☒ ~~Decision 2: which envelope~~ — §9. Derated-joint-4, at the §9.3 pose.
- ☐ **Human:** decision 3 (the framing, §10.4) — now the highest-stakes one.
- ☐ **Human:** decision 1, which §9.3 suggests is moot unless the application pins the pose.
- ☒ ~~Root-cause the three Section 7 findings~~ — §8.1. One constraint, monotone in K_0 .
- ☒ ~~Fix the precondition failure~~ — §8.2, `nominal_auth_mono`. Both guarantees to $16\times$.
- ☒ ~~Try anisotropic (K_0, D_0)~~ — §8.3. It does not help here and slightly hurts; the displacement is 8:1 push-axis dominated.
- ☒ ~~Re-run the gate at a pose that does not load joint 4~~ — §9.3. 481 cells against 33, and 45.7 % headroom against 2.4 %.
- ☒ ~~Re-run Tasks 2 and 3 at the §9.3 pose~~ — §9.4. Both axes are now exercised by `phri2`’s own benchmark, and §7.5’s “flatter” conclusion is corrected: the trade is 7.7 %, as the synthesis note predicted.
- ☐ Re-run §8’s root-cause and fix comparison at the recommended pose too. They are diagnoses of a problem the pose largely removes, so they are still correct as history, but the numbers a paper quotes should be the recommended pose’s.
- ☐ Search poses properly rather than on a q_2, q_4, q_6 grid screened by one scenario, and check whether the recommended pose is one a real task would accept.
- ☒ ~~Joint pose $\times K_0$ scan looking for a well-behaved dual-axis operating point~~ — §10.5. 96 cells; one candidate, conditional on the box reading.
- ☒ ~~Promote §10.5’s candidate to a proper case study — repeat it across seeds and disturbance profiles~~ — §10.6. It does not survive: 5/20 on all certificates, and the 16 % excursion was a mid-range draw of a 100-163 % spread. Demoted to a remark.
- ☒ ~~Restate Merged Lemma 1 to say what is guaranteed when the precondition fails but α_{nom} is active~~ — §8.5, Merged Lemma 1’. Checked clause by clause on 4000 adversarial ticks. It also produced two results that were not being looked for: (P3) is what makes the energy half provable at all, and (C4)’s exchange rate re-opens decision 0.

- ✓ ~~Implement the merged controller and re-measure rows 1 and 4 in its own closed loop~~ — §7.1–7.2. Replay and closed loop agree to 0.1 %.
- ✓ ~~Task 3: re-sweep the authorization vs tracking curve (E_0) with both tightenings active~~ — §7.4–7.5. The trade is **flatter**, not steeper, and §7.5 says why that is a bad sign rather than a good one.
- **Task 2, the proof:** write merged Lemma 1 (per-joint α_r closed form, $\alpha_r \cdot \alpha_E$ on-segment lemma) and merged Proposition 1 at submittable granularity. The implementation supplies the operating point and confirms both halves empirically; the write-up must state the precondition as a **standing hypothesis that can fail at run time** (§7.7), not as a design-time check.
- ✓ ~~Decide what the controller does when the precondition fails~~ — §8.2. Folding the nominal into the authorization loop works, and the passivity cost it first appeared to carry is removed by making α_{nom} monotone. The escalation route back into certified-realizability is still the answer for the residual condition §8.4 names (τ_{base} itself), which no local fix defends.
- Re-run the Task 1 gate at a pose/push direction that does not load joint 4, and check whether the headroom problem is FR3-pose-specific.
- **Implement and test the per-contact reset** `nominal_auth_mono` needs. §8.5's (C5) upgrades this from a nicety to a correctness requirement: without it (C3)'s inertness is empty beyond the first firing.
- **State what the derated envelope means** — thermal/duty-cycle budget or safety-certified limit. §8.5's (C4) makes decision 0's trade turn on it, and it is not answerable from inside the simulation.
- **Resolve whether the 0.06 m bound is hard or slack-relaxed.** Back to being a Task 1 question only — it decides a 33-cell versus 6-cell region. §10.5 briefly made it decisive for the framing too; §10.6 retired that, because the exhibit it would have licensed does not survive resampling.
- Explicitly disclaim the force-misclassification pillar in whatever is written. §7.1's channel split makes the assumption visible; it does not discharge it.

13. Reproducing

```

cd simulation
python3 pir_knot_scan.py                # ~19 min on 4 cores (Task 1 gate)
python3 pir_knot_scan.py --rescore      # re-derive verdicts/figures from stored maps
python3 pir_verify.py                  # ~3 min (gate verification)
python3 run_pir_closed_loop.py --scenario push # ~35 s (Section 7.2, 7.6)
python3 run_pir_closed_loop.py --scenario merged # ~35 s (Section 7.3, 7.4)
python3 pir_e0_sweep.py                # ~6 min (Section 7.4, 7.5)
python3 pir_rootcause.py               # ~4 min (Section 8.1)
python3 pir_fixes.py                   # ~5 min (Section 8.3)
python3 pir_pose_study.py              # ~4 min (Section 9.1, 9.2)
python3 pir_knot_scan.py --pose -1.30 -1.30 1.571 \
    --tag pose_q2m13_q4m13             # ~19 min (Section 9.3)
POSE="--pose -1.30 -1.30 1.571 --k0 520.0 --d0 56.13 --tag pose_q2m13_q4m13"
python3 run_pir_closed_loop.py $POSE --scenario push # (Section 9.4)
python3 run_pir_closed_loop.py $POSE --scenario merged # (Section 9.4)
python3 pir_e0_sweep.py $POSE          # ~25 min (Section 9.4)
python3 pir_rootcause.py --pose -1.30 -1.30 1.571 --tag pose_q2m13_q4m13
python3 pir_fixes.py --pose -1.30 -1.30 1.571 --k0 520.0 --d0 56.13 \
    --tag pose_q2m13_q4m13             # ~9 min (Section 8)
python3 pir_axis_tension.py            # seconds; reads results/ (Section 10)
python3 pir_joint_scan.py              # ~8 min (Section 10.5)
python3 pir_robustness.py              # ~10 min (Section 10.6)
python3 pir_lemma_check.py             # ~2 min (Section 8.5)
python3 -m pytest test_pir_knot_scan.py test_pir_controller.py -q

```

MuJoCo mesh assets are gitignored repo-wide and must be fetched from MuJoCo Menagerie into

```

../simulation/models/franka_fr3/assets/ . Pinned dependency versions are in ../impedance/simulation/requirements.txt ;
mujoco==3.10.0 matters, since 3.13 renamed MjData.qm and fr3_mujoco.py reads it.

```